

Conformational dynamics and asymmetry in multimodal inhibition of membrane-bound pyrophosphatases

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The current study presents **useful** findings about the inhibition of a membrane pyrophosphatase by non-hydrolyzable phosphonate substrate analogs. The study proposes a model in which the two monomers in a functional dimer interact with the phosphonate molecules in an asymmetric fashion. While asymmetry has been previously demonstrated through other studies, the DEER spectroscopy data presented in the current study provide **incomplete** evidence of the proposed asymmetry near the binding site.

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Abstract

Membrane-bound pyrophosphatases (mPPases) are homodimeric proteins that hydrolyse pyrophosphate and pump H^+/Na^+ across membranes. They are crucial for the virulence of protist pathogens, making them attractive drug targets. In this study, we investigate the inhibitory effects of seven distinct bisphosphonates against *Thermotoga maritima* mPPase to explore their mode of action and assist in future small molecule inhibitor development. We solved two structures of mPPase bound to the inhibitors in the enzyme active sites and probed the conformational dynamics of mPPase under multiple inhibitors and functionally relevant conditions by double electron-electron resonance (DEER) spectroscopy. We found that mPPase adopts five distinct conformations within the ensemble in the presence of different inhibitors. Combined with solid-supported membrane-based electrophysiology recordings, this revealed that during catalysis, one monomer of the dimer remains open, and

Na^+ can only be pumped in a closed state. These results further support the existence of catalytic asymmetry in our pumping-before-hydrolysis model.

Introduction

Membrane-bound pyrophosphatases (mPPases) facilitate the transport of protons and/or sodium ions across membranes while catalysing the breakdown of pyrophosphate (PP_i), a by-product generated in various cellular synthetic reactions - into inorganic phosphate (P_i). These enzymes are found in plants, certain species of bacteria, protist parasites, and archaea, but are absent from multicellular animals^{1,2,3,4,5}. Within these organisms, mPPases are essential for cell survival under diverse stress conditions such as osmotic stress, mineral deficiency, and extreme temperature⁶. Based on their potassium dependency, mPPases are divided into two families: K^+ -dependent and K^+ -independent. While K^+ -independent mPPases all transport H^+ , K^+ -dependent mPPases can transport H^+ , Na^+ , or both⁴.

Currently, mPPase structures from three different organisms have been reported: *Vigna radiata* (VrPPase), *Thermotoga maritima* (TmPPase), and, most recently, a structure from *Pyrobaculum aerophilum* (PaPPase) in complex with imidodiphosphate (IDP)⁷. For TmPPase, several different structural states have been determined, including the resting state (TmPPase:Ca:Mg)⁸, with two phosphates bound (TmPPase:2 P_i)⁸, IDP bound (TmPPase:IDP)⁹, IDP and *N*-(2-amino-6-benzothiazolyl)methyl]-1*H*-indole-2-carboxamide (ATC) bound (TmPPase:IDP:ATC)¹⁰, phosphate analogue (WO_4)-bound (TmPPase: WO_4)⁹, and time-resolved X-ray diffraction structures (with and without substrate/product bound) showing structural asymmetry⁷. Similarly, VrPPase has been solved in multiple states, including IDP-bound (VrPPase:IDP)¹¹, single phosphate-bound (VrPPase: P_i)⁹, two phosphates bound (VrPPase:2 P_i), and different mutations at the hydrophobic gate¹². These structures show that mPPases are homodimeric enzymes, with each monomer consisting of 16 transmembrane helices (TMHs), occasionally 17 TMHs as found in the sequence databases, organised into two concentric rings: the inner ring (TMH5-6, 11-12, and 15-16) and the outer ring (TMH1-4, 7-10, and 13-14). Each monomer consists of four regions: a hydrolytic centre, a coupling funnel, an ion gate, and an exit channel¹³ (**Fig. 1A**). To simplify residue comparison between mPPases, we employ the residue numbering scheme $\text{X}\Sigma^{\text{YZ}}$ (superscripts refer to Ballesteros–Weinstein numbering¹⁴), where X represents the amino acid, Σ denotes the amino acid position in TmPPase, Y indicates the helix number, and Z specifies the offset of amino acid positions within the centrally conserved residues of the helix^{12,15}.

mPPases are a promising drug target for treating diseases caused by parasitic protists, such as malaria and leishmaniasis^{10,16,17,18}. Among the currently available compounds, ATC demonstrates the most effective inhibitory activity against TmPPase²⁰. ATC is bound to a region near the enzyme exit channel of one subunit, which induces structural asymmetry in the mPPase dimer²⁰. Functional asymmetry in K^+ -dependent mPPases has also previously been shown by Artukka, et al.²¹. Anashkin and coworkers²² further supported this hypothesis by analysing the inhibition of *Desulfitobacterium hafniense* mPPase using three non-hydrolysable PP_i analogues (IDP, etidronate (ETD), and aminomethane bisphosphonate). Bisphosphonates, such as risedronate (RSD) and pamidronate (PMD), serve as primary drugs currently used to combat osteoclast-mediated bone loss²³. Unlike IDP, which contains a P-N-P bond, bisphosphonates have a P-C-P bond, with its central carbon can accommodate up to two substituents, allowing a large compound variability. Therefore, understanding their inhibition mechanism on mPPases is crucial for developing future small molecule inhibitors.

Our previous work on serial time-resolved X-ray crystallography and electrometric studies on TmPPase showed a direct observation of structural asymmetry, where two monomers are in different states during PP_i hydrolysis upon the addition of substrate and Na^+ , supporting a “pumping-before-hydrolysis” energy coupling model⁷. However, except for the allosteric

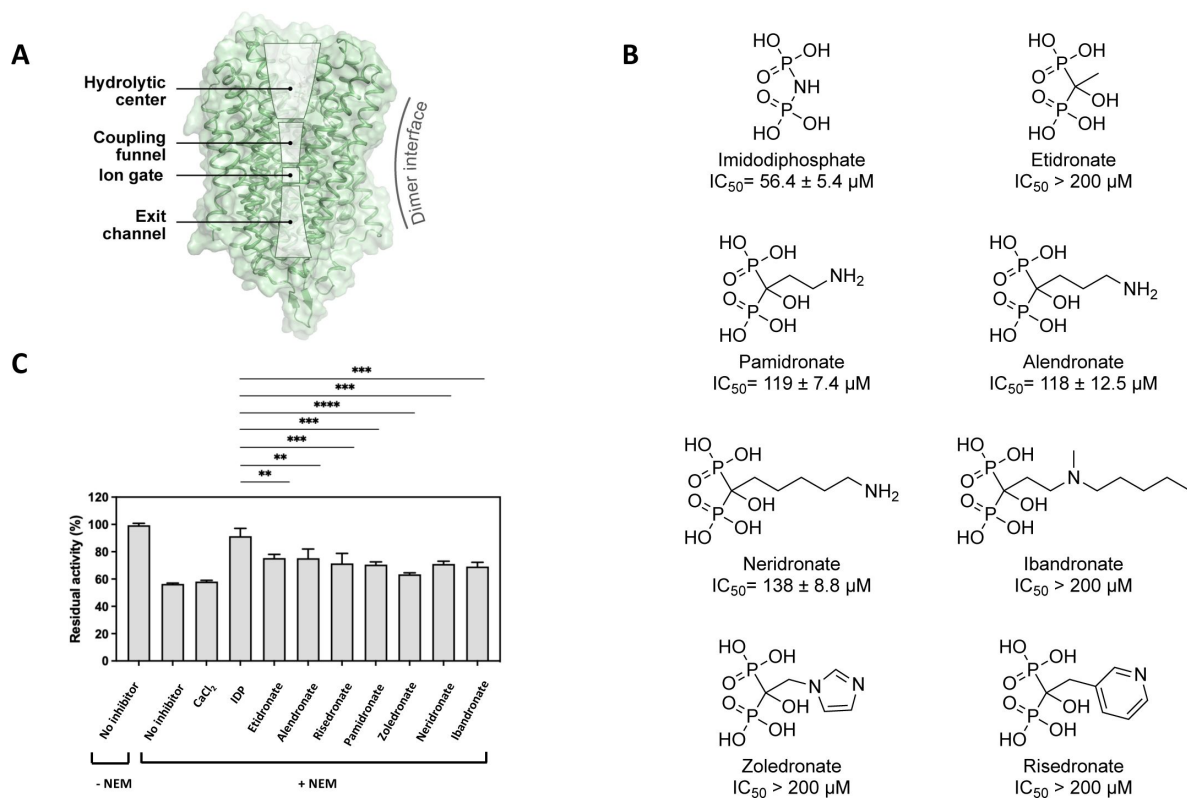


Figure 1.

Inhibition effect of bisphosphonates on TmPPase.

A. Overall structure of the monomer TmPPase structure, consisting of hydrolytic centre, coupling funnel, ion gate and exit channel. **B.** Chemical structure of IDP and bisphosphonates and their inhibition activity against TmPPase. **C.** Activity of TmPPase modified with NEM (N-Ethylmaleimide) in the presence of MgCl₂ (2.5 mM) and NaCl (20 mM) after incubation with Ca²⁺ (2 mM) and bisphosphonates (0.5 mM). Measurement was done in triplicate.

inhibitor ATC, which binds to a region near the exit channel, crystal structures of TmPPase bound to inhibitors at the active site are symmetric. In order to probe the proposed asymmetry caused by the inhibitor (and substrate) binding in solution, we employed double electron-electron resonance (DEER), also known as pulsed electron double resonance (PELDOR) spectroscopy. This method relies on the introduction of spin labels on selected protein sites, allowing for precise determination of inter-spin distances²⁴⁻²⁶, making it a powerful tool for probing the conformation and dynamics of integral membrane proteins²⁷⁻³⁰, including ion channels, transporters, outer membrane proteins, and receptors in their native environments³¹⁻³⁷. As an ensemble technique, DEER can probe the presence of multiple conformational species in solution including lowly populated protein states, which are key to the protein function³⁸⁻⁴⁰. Here, we solved two TmPPase structures in complex with ETD and zoledronate (ZLD) and monitored their conformational ensemble using DEER spectroscopy in solution. Overall, bisphosphonates can trigger conformational changes in the active site and near the exit channel of TmPPase in an asymmetric mode and under certain inhibitor-bound conditions, the DEER data correlate with the expected distances of an open/closed asymmetric model, consistent with the corresponding X-ray structures. This, along with our electrometric studies detecting the Na⁺ signal across the membrane, further suggests that ion pumping requires a fully closed state of one TmPPase monomer, supporting our pumping-before-hydrolysis model⁷.

Results

Bisphosphonates are weaker TmPPase inhibitors than IDP

Bisphosphonates have been shown to inhibit mPPases^{22,41}. To understand their binding mechanism to TmPPase, we first assessed the binding ability of seven distinct bisphosphonates to TmPPase by testing their inhibitory activity using the molybdenum blue assay⁴² (**Fig. 1B**), with IDP (IC₅₀ = 56.4 ± 5.4 μM) as a positive control¹⁹. Of the compounds tested, all the straight-chain primary amines had similar IC₅₀s, ranging from 117 to 138 μM (*P* = 0.06). Substituting the -NH- of IDP with the -CCH₃(OH)- of ETD resulted in a significantly weaker IC₅₀ (> 200 μM). Similarly, branched aliphatic and aromatic bisphosphonates (ibandronate (IBD), ZLD and RSD) also showed weaker inhibition (IC₅₀ > 200 μM) (**Fig. 1B** and **Fig. EV1**).

To confirm that the binding of bisphosphonates to TmPPase induces conformational changes in the protein structure, we incubated the enzyme with the inhibitors and performed an *N*-ethyl maleimide (NEM) modification assay⁴³. NEM covalently binds to exposed cysteine residues of the protein, forming a carbon-sulfur bond that can inhibit the protein activity if the residue is essential⁸. IDP has been reported to prevent the NEM modification of cysteine by retaining TmPPase activity⁸. In the absence of inhibitors, NEM modification resulted in a decrease in TmPPase activity by approximately 40% (**Fig. 1C**), similar to the activity reduction observed with CaCl₂, an inhibitor that binds to the open form of TmPPase⁸. Upon the addition of IDP, TmPPase adopts a closed conformation, rendering it resistant to NEM modification⁸ (**Fig. EV2**); consequently, the enzyme remains largely unaffected by NEM. Although not as effective as IDP, all bisphosphonates prevent NEM modification to a comparable extent (**Fig. 1C**).

TmPPase structures in complex with bisphosphonate inhibitors

To decipher the structural basis of bisphosphonates inhibition and their binding to TmPPase, we decided to solve their structures since all the bisphosphonates bound to TmPPase despite not being isosteres of PP_i (**Fig. 1**). We obtained protein crystals for all the inhibitors, but they diffracted weakly, except for TmPPase in complex with ETD (TmPPase:ETD) and ZLD (TmPPase:ZLD), which diffracted to resolutions of 3.2 Å and 3.3 Å, respectively. TmPPase:ETD crystallised in the presence of Ca²⁺, which is a well-known mPPase inhibitor⁸, while TmPPase:ZLD crystallised without Ca²⁺. Both data sets were anisotropic as analysed using the STARANISO server⁴⁴ (**Table S1**). We solved both structures by molecular replacement using

the resting state structure (PDB ID: 4AV3) as the search model for TmPPase:ETD and the closed IDP-bound structure (PDB ID: 5LZQ) for TmPPase:ZLD. There were two molecules in the asymmetric unit for TmPPase:ETD and four for TmPPase:ZLD.

In the initial round of the refinement for the TmPPase:ETD structure, both chains displayed positive ($F_o - F_c$) density at 3σ in their hydrolytic centres that could accommodate ETD (**Figs. EV3A-B**, [upper left panel](#)). We also observed extra density that corresponds to a calcium ion in the resting state structure⁸ (**Figs. EV3A,B**, [upper left panel](#)). Due to the high Ca^{2+} concentration (0.2 M) in the crystallisation condition, we placed the same ion at this position. After placing Mg^{2+} ions and water molecules in the difference density peaks, further rounds of refinement provided us with a reasonable $2mF_o - F_c$ density map in the active site of both monomers (**Fig. EV3A,B**, [right panel](#)) and the POLDER (Omit) maps indicate a good fit of the compound to the density (**Fig. EV3A,B**, [bottom left panel](#)). Finally, the TmPPase:ETD structure was refined to an average resolution of 3.2 Å ($h = 3.1$ Å, $k = 3.6$ Å, $l = 4.3$ Å) with the final $R_{\text{work}}/R_{\text{free}}$ of 27.2% / 31.0 % (**Table S1**).

Similarly, the initial refinement of TmPPase:ZLD revealed positive ($F_o - F_c$) density at 3σ that could accommodate ZLD in all four chains in the asymmetric unit (**Figs. EV4A-D**, [upper left panel](#)). After placing Mg^{2+} ions and water molecules in the difference density peaks, further rounds of refinement provided us with a $2mF_o - F_c$ density map in the active site for all monomers (**Fig. EV4A-D**, [right panel](#)) and was validated by POLDER (Omit) maps (**Figs. EV4A-D**, [bottom left panel](#)). The final refinement shows that the TmPPase:ZLD structure has an average resolution of 3.3 Å ($h = 4.5$ Å, $k = 4.2$ Å, $l = 3.2$ Å) with a final $R_{\text{work}}/R_{\text{free}}$ of 25.9 % / 30.4 % (**Table S1**).

Asymmetry in the TmPPase complex with etidronate

Unlike the fully-open TmPPase:Ca:Mg structure (PDB ID: 4AV3), we observed additional density above the hydrolytic centre in both chains that could be fitted with several residues of loop5-6 (**Fig. 2A** and **Fig. EV5**). This left eight residues (V208^{5.67}-L215^{5.74}) in loop5-6 of chain A and three residues (L213^{5.72}-L215^{5.74}) in chain B unmodeled due to the lack of extra density. In the IDP-bound structure⁴⁵, these loops interact with IDP and form a tightly packed structured lid over the active site. However, in the TmPPase:ETD structure, despite interacting with ETD in both chains, these loops are positioned slightly above the active site (**Fig. 2B-D**), with loop5-6 of chain A extending more toward the centre compared to loop5-6 of chain B (**Fig. 2A**).

Structural alignment between chain A and B of TmPPase:ETD yields a root mean square deviation (RMSD) per Ca of 1.44 Å, approximately four times higher than the RMSD between chain A and B in the resting state (RMSD/Ca = 0.39 Å) (**Table S2**). Despite the overall structural similarity, further comparison between the monomers of TmPPase:ETD and those in the resting state reveals that chain B of TmPPase:ETD differs most from TmPPase:ETD chain A and from both chains in the resting state structure (**Table S2**). Notably, there are clear differences on the cytoplasmic (hydrolytic) side between the monomers of TmPPase:ETD; chain B adopts a slightly more constricted conformation compared to chain A (**Fig. 2A**). Bendix analysis⁴⁶ showed that three (TMH11, 12 and 15) out of six inner ring helices of chain B are more curved on the cytoplasmic side, bending towards the active site (**Fig. EV6**). Besides that, the loops on the cytoplasmic side of chain B (loops11-12, 13-14, and 15-16) appear to be more flexible, as indicated by more unresolved residues, compared to those in chain A (**Fig. 2A**). These observations suggest structural asymmetry between chains A and B in the TmPPase:ETD structure.

The structural asymmetry arises due to differences in the orientation of ETD within each monomer at the active site. ETD comprises two phosphonate groups separated by a central carbon bonded to a hydroxyl group. Compared to the IDP location in the IDP-bound structure, ETDs are positioned above the IDP site, with the lower phosphonate group of ETDs located in the position of the upper (leaving-group) phosphonate group of IDP (**Fig. 2B**). However, the upper phosphonate group of ETDs in chains A (ETD_A) and B (ETD_B) is distinctly positioned; ETD_A is tilted

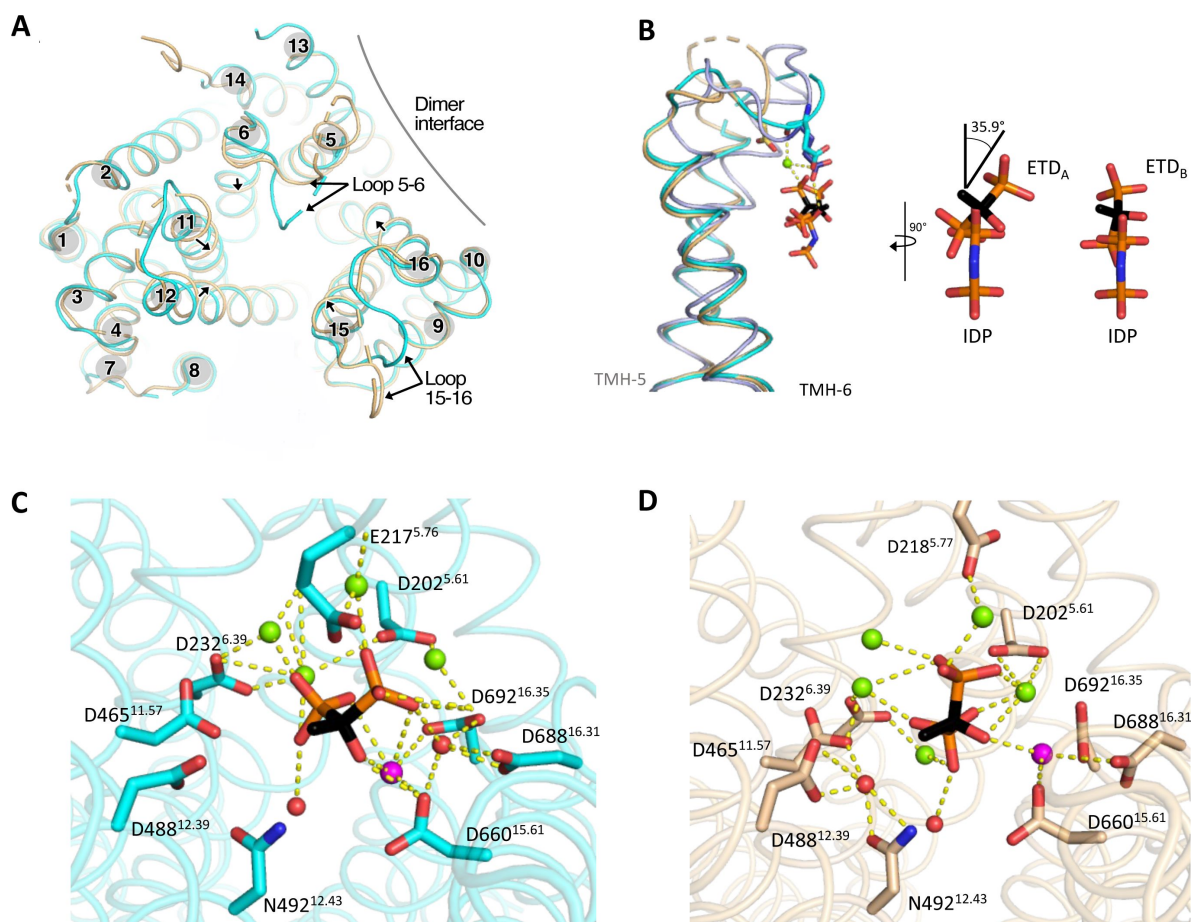


Figure 2.

Structural asymmetry in the dimer active-site of TmPPase:ETD complex.

A. Superposition of chain A (cyan) and chain B (wheat) showing the relative movements (black arrow) of helices. **B.** Superposition of TMH5 and TMH6 in TmPPase:ETDs and TmPPase:IDP (orange; PDB: 5LZQ) showing the movement of loop5-6 and reorientation of ETD_A, ETD_B and IDP. The dashed line shows the interaction of E217^{5.76} of loop5-6 in Chain A with ETD_A and IDP, and D218^{5.77} of loop5-6 in Chain B with ETD_B; Close-up view of IDP superposed with ETD_A and ETD_B. **C.** Residues in the active site with ETD_A coordinated (dashed lines), Ca²⁺ (pink sphere) and nucleophilic water (red spheres) in a Mg²⁺ metal cage (green spheres). **D.** Residues in active site with ETD_B coordinated (dashed lines), Ca²⁺ (pink sphere) and water (red spheres) in a Mg²⁺ metal cage (green spheres).

approximately by 35.9° relative to the IDP orientation, while ETD_B is relatively parallel to the IDP orientation (**Fig. 2B**). The lower phosphonate group position remains the same for both ETDs (**Fig. 2B**). As a result, loop5-6 of the two monomers is oriented differently. In chain A, this loop protrudes towards the active centre and interacts with ETD_A via E217^{5,76} mediated by a Mg²⁺ ion, while in chain B, the loop is more constricted and interacts with ETD_B via D218^{5,77}, also mediated by a Mg²⁺ ion (**Fig. 2B,C** and **D**). Furthermore, ETD_A and ETD_B interact with the active site via different residues (**Fig. 2C,D**). D465^{11,57}, D488^{12,39} and N492^{12,43} in TMH11 and TMH12 were involved in the interaction with ETD_B via a water molecule. Consequently, these two TMHs undergo slight inward movement, resulting in a more constricted conformation of chain B. Nonetheless, the methyl group of ETDs in both chains points towards TMH12 (**Fig. 2C,D**), which might prevent complete closure of the hydrolytic centre and downward motion of TMH12.

Structural distinction between zoledronate and IDP-bound TmPPase

In contrast to the TmPPase:ETD structure, the TmPPase:ZLD structure adopts a closed conformation. The overall structure is more similar to the IDP-bound structure (RMSD/Cα of 0.760 Å) than the resting state structure (RMSD/Cα of 2.32 Å) (Table. S2). However, compared to the IDP-bound structure, the TmPPase:ZLD structure exhibits noticeable movements in three of six inner ring helices (TMH11, 12 and 15) and seven of ten outer ring helices (TMH1-4 and 7-9). These movements extend outwards from the hydrolytic centre (**Fig. 3A**), leaving it not fully closed. A cross-sectional view confirms this observation, showing the tunnel extending from the hydrolytic centre to the enzyme opening, which is closed in the IDP-bound structure (**Fig. 3B,C**). This is because ZLD is bulkier than IDP due to the presence of the heteroaryl group, which points towards TMH11, 12, and 15 on the cytoplasmic side.

Although the hydrolytic centre of TmPPase:ZLD is more open, the coordination of the Mg₄ZLD complex with the active site residues closely resembles that of Mg₅IDP in the IDP-bound structure (**Fig. 3E,F**). ZLD is nonetheless positioned about 1.0 Å above IDP (**Fig. 3D**) because it is bulkier. However, unlike the IDP-bound structure, and, even though the arrangement of TMHs in the ion gate is almost identical, we did not observe any density for a Na⁺ in the TmPPase:ZLD structure despite its higher resolution (*i.e.* 3.26 Å compared to 3.5 Å for the IDP-bound structure) (**Fig. EV7C**).

Exit channel and active site lid movements

The X-ray structures of TmPPase with the different inhibitors bound to the active site show either a closed, resting or asymmetric (TmPPase:ETD, **Fig. 2**) conformation. The asymmetric structure of the TmPPase with ETD is unique and similar to that observed in our recent time-resolved study⁷. To probe the TmPPase conformational ensemble in solution under various inhibitor-bound conditions, we employed DEER spectroscopy. We selected two distinctive sites on TmPPase (S525 and T211), which were mutated to cysteine and selectively spin-labelled with 2,5-dihydro-2,2,5,5-tetramethyl-3-[[[(methylsulfonyl)thio]methyl]-1*H*-pyrrol-1-yloxy (MTSSL, modification is denoted as R1 hereafter) to enable the measurement of interspin distances between the spin-labelled residue pairs. The selected sites were located on the exit channel (S525R1) and the enzyme entrance (T211R1) to capture any conformational changes occurring in the protein on either side of the membrane (**Fig. 4A** and **Fig. EV8A**) without interfering with the activity of TmPPase. For both TmPPase sites, high mPPase spin labelling efficiency was achieved with no non-specific MTSSL being present, as evidenced by the continuous wave electron paramagnetic resonance (CW EPR) spectra recorded at room temperature (**Fig. EV9**)⁴⁷⁻⁴⁹. From the CW EPR spectral lineshapes (**Fig. EV9**), which relate to the rotation correlation time of the spin label, we observed that T211R1 showed high spin label mobility in the conditions tested (apo, +ETD, and +IDP), while S525R1 had intermediate to low spin label mobility in the conditions tested (apo, +Ca, +Ca/ETD and +ETD). This is consistent with spin-labelled sites on integral membrane proteins of variable dynamics³⁰⁻³². Unlike DEER, which provides insights into the conformational state of

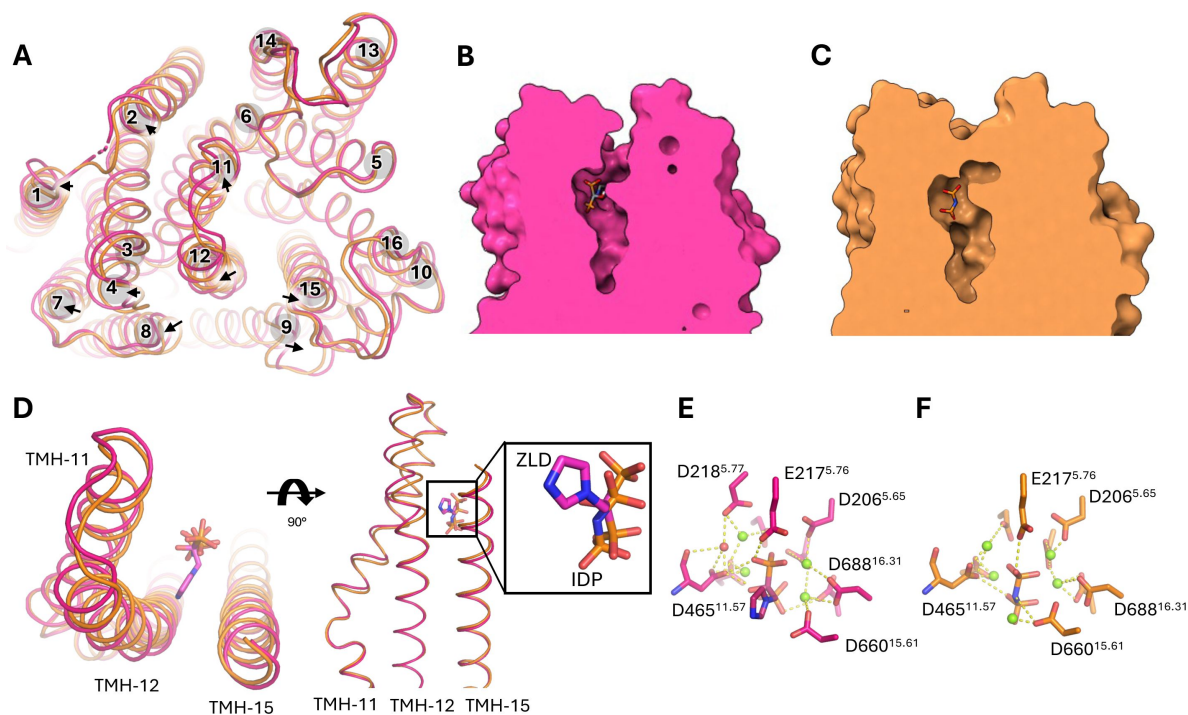


Figure 3.

Comparison of the TmPPase:ZLD and TmPPase:IDP structures in the active site.

A. Superposition of TmPPase:ZLD (chain A, pink) and TmPPase:IDP (chain A, orange) (PDB: 5LZQ). Helices movements are indicated by a black arrow. **B** Cross-section view of the active site in TmPPase:ZLD. **C.** Cross-section view of the active site in TmPPase:IDP. **D.** Superposition of TMH11, TMH12 and TMH15 in TmPPase:ZLD and TmPPase:IDP showed the movement of the hydrolytic centre and the orientation of ZLD and IDP. **E.** The coordination of key residues in the active site with ZLD (dash line), and water (red sphere) in a Mg^{2+} metal cage (green spheres). **F.** The coordination of key residues in the active site with IDP (dash line) in a Mg^{2+} metal cage (green spheres).

membrane proteins³⁰, CW EPR offers information on the local environment of the spin label. The results show no significant difference in the local environment at either site between the *apo* state and the inhibitor-bound state.

In addition, we generated *in silico* distance distribution predictions for the two sites (S525R1 and T211R1) using MtsslWizard⁵⁰ and based on the X-ray structures of TmPPase with different molecules bound (**Fig. 4**, **Fig. EV8**). In the case of T211R1, the X-ray electron density in loop_A⁵⁻⁶ of the TmPPase:ETD (residues V208^{5.67}-L215^{5.74}: VGKTELNL) and TmPPase:Ca (residues T211^{5.70}-R221^{6.28}: TELNLPEDDPR) structures is missing, suggesting a highly dynamic or disordered state for this region. We therefore modelled this protein region using the Rosetta server⁵¹ and generated *in silico* distance distributions based on the X-ray structures. These were overlaid with the experimentally derived DEER distance distributions (**Fig. 4D**, **Fig. EV8D**) for comparison. The S525R1 dimer yielded high-quality DEER traces, allowing us to resolve the distance differences between the inhibitor conditions probed. Under all tested conditions, strong oscillations were observed in the raw DEER data (**Fig. 4B**) indicating that the shifts observed in the state populations within the TmPPase ensemble are highly reliable. Both DeerAnalysis and ComparativeDeerAnalyser were used for background correction of the DEER traces and their results are in agreement (**Fig. 4C** and **Fig. EV8B**).

The separation of S525R1 in the *apo* state (with no Ca²⁺ or inhibitor added) is broad with a mean distance of 3.8 nm (**Fig. 4D**). In the presence of Ca²⁺, the distance distribution shifts towards shorter distances, suggesting that the two monomers come closer at the periplasmic side, and consistent with the predicted distances derived from the TmPPase:Ca structure. In the presence of both Ca²⁺ and ETD (+Ca/ETD), we observe an increase in the mean distance distribution to 3.7 nm, consistent with the predicted distance for the TmPPase:ETD structure (which corresponds to the +Ca/ETD condition). Finally, in the presence of ETD but no Ca²⁺, the experimental distance between the loops on the different monomers increases to 4.0 nm. Although these changes are small, they are significant. They can be seen in both the time domain data (**Fig. 4C**) and resulting distances (**Fig. 4D**). The reference black dashed lines depicted for the *apo* state, highlight these subtle changes in the background corrected time traces. More substantial changes occur with the addition of IDP with a mean of the resulting distance distribution at 4.2 nm, which is the longest distance recorded for the S525R1 pair in solution. In contrast, the addition of ZLD results in the shortest distances we observed by DEER under the conditions tested, with a mean of the distance distribution of 3.4 nm. Remarkably, this differs substantially from the *in silico* distance distribution predicted from the X-ray structure of TmPPase:ZLD (4.3 nm), which is expected to be identical to that of TmPPase:IDP (RMSD/Ca = 0.571 Å) (see Discussion).

Unlike S525R1, distance distributions for T211R1 derived from the DEER time traces are broad for all conditions tested (**Fig. EV8D**). This is evidenced by the lack of oscillations in the raw DEER data (**Fig. EV8B**), which further supports the highly dynamic nature of the labelling site located on loop_A⁵⁻⁶. The lack of electron density in the respective X-ray structures is due to flexibility, also consistent with the high mobility recorded in our CW EPR spectra at room temperature (**Fig. EV9A**). DeerAnalysis and ComparativeDeerAnalyzer are in close agreement, as in the case of S525R1 (**Fig. EV10**). For *apo* and +Ca/ETD in particular, traces were collected for longer time windows to resolve longer distances for the T211R1 pair (**Fig. EV8B**). The broad distance distributions suggest the conformational flexibility of T211R1 (**Fig. EV8C**). The *apo*, +Ca and +Ca/ETD conditions all appear highly similar, while the obtained distributions for +ETD and +IDP have larger contributions from shorter distances within their conformational ensembles. Finally, TmPPase incubated with ZLD (TmPPase+ZLD) in solution appears to have longer distances, contrasting with predictions derived from the TmPPase:ZLD structure.

Collectively, periplasmic S525R1 allows for robust quantitative interpretations of the conformational transitions TmPPase undergoes under different conditions. T211R1 presents a highly dynamic site with less defined oscillations in the DEER traces. However, the overall

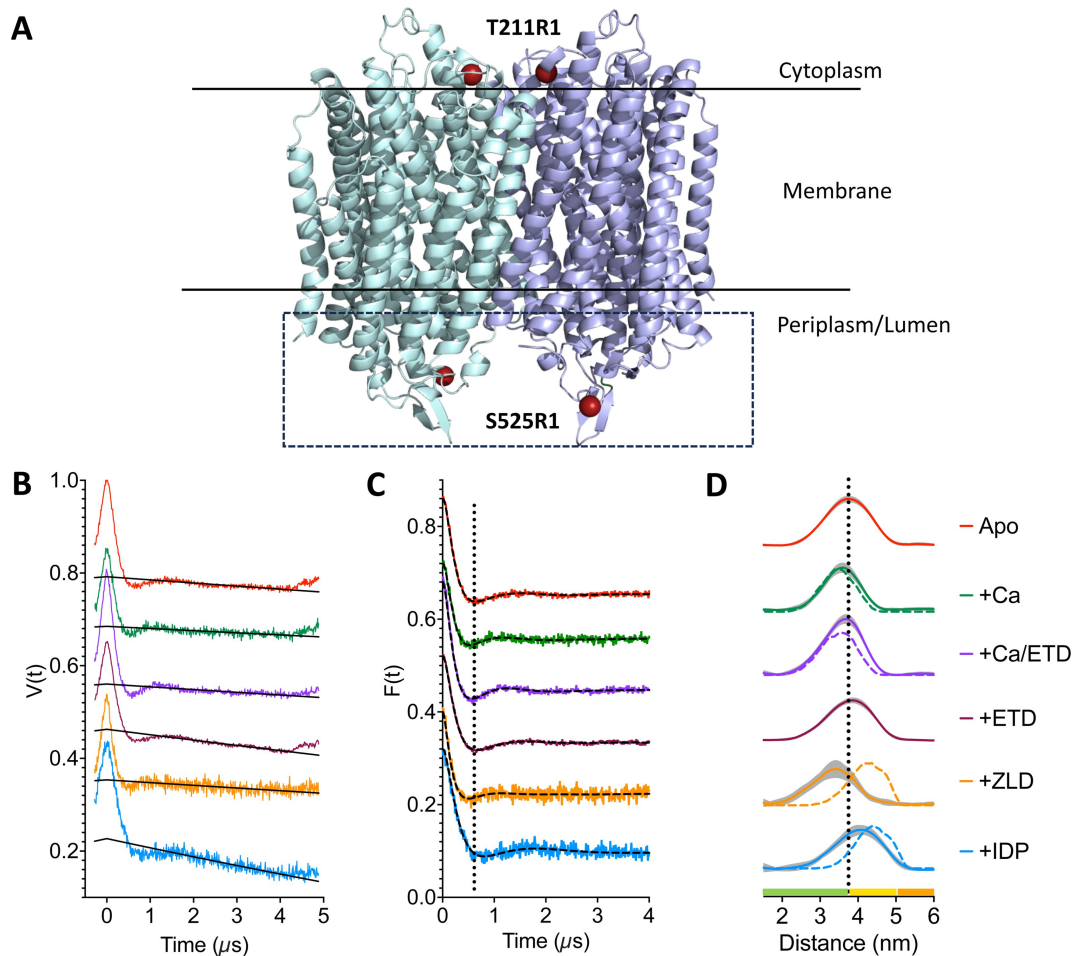


Figure 4.

DEER distance distributions of TmPPase S525R1 under different conditions.

A. Structure of the TmPPase dimer, with monomers A and B coloured cyan and purple, respectively. The sites mutated to cysteine and labelled with MTSSL are shown by maroon spheres, with T211R1 on the cytoplasmic (top) side and S525R1 on the periplasmic (bottom) side of the membrane. **B.** DEER raw data traces for S525R1. Each condition measured is coloured accordingly. **C.** DEER background corrected time-domain traces for S525R1. The black dashed line represents the base of the oscillation in the *apo* state and acts as a guide to highlight the small distance changes where the base of the oscillation comes before (shorter distance) or after (longer distance) the line. **D.** The overlap between the predicted distance distribution of S525R1 from the solved crystal structures (TmPPase:Ca, TmPPase:Ca:ETD, TmPPase:ZLD, and TmPPase:IDP), shown as coloured dashed lines, with the resulting DEER distance distributions at the respective conditions. The vertical black dashed line represents the mean distance for the *apo* state to highlight changes from this to shorter and longer distances. The grey-shaded regions represent the uncertainty in the distribution and the coloured bar above the x-axis represents the DeerAnalysis 'traffic light' system for assessing the reliability of the distribution presented. For the time window of this dataset, the distances within the regions highlighted in green and yellow indicate the reliability of the mean distance, peak shape, and width. For distance peaks within the orange region (> 5.0 nm), only the mean distance is considered reliable⁷⁰. The data were all processed in DeerAnalysis, with validation in the same way as described in the methods.

transitions suggested by T211R1 agree qualitatively with conformational changes informed by the S525R1 pair: upon substrate/inhibitor binding, the T211R1 pair moves closer while the S525R1 pair distance increases.

Based on the DEER data on the IDP-bound TmPPase, we observed significant deviations between the experimental and the *in silico* distances derived from the TmPPase:IDP X-ray structure for both cytoplasmic- (T211R1) and periplasmic-end (S525R1) sites (**Fig. 4D** [↗](#) and **Fig. EV8D** [↗](#)). This deviation could be explained by the dimer adopting an asymmetric conformation under the physiological conditions used for DEER, with one monomer in a closed state and the other in an open state. To model this asymmetric arrangement, we combined chain A of the TmPPase:Ca structure with chain B of the TmPPase:IDP structure, *i.e.* TmPPase:Ca(A)_IDP(B), and predicted its *in silico* distance distribution (**Fig. 5A** [↗](#)). Remarkably, the derived asymmetric model predicts a distance distribution that agrees closely with the experimental DEER distance distribution obtained for S525R in solution (**Fig. 5B** [↗](#)). This predicted distribution also falls between the two conformational extremes of TmPPase (**Fig. 4D** [↗](#) and **5B** [↗](#)).

Effect of ETD and ZLD on sodium transport of TmPPase

Previously, we showed that IDP can facilitate a single Na^+ pumping cycle without hydrolysis⁷ [↗](#). To investigate whether pumping also occurs in the presence of ETD and ZLD, we recorded electrometric data during PP_i hydrolysis and after binding of IDP, ETD and ZLD. In electrometric measurements, also known as solid-supported membrane-based electrophysiology⁵² [↗](#), a current signal is generated and recorded when Na^+ is transported across the membrane by the active reconstituted TmPPase. A maximal positive signal of 0.6 ± 0.025 nA was detected within 0.15 ns (excluding instrument dead time) after the addition of 100 μM substrate K_4PP_i (**Fig. 6A** [↗](#)). Most of the signal decayed within 1 second after K_4PP_i was supplied. Full signal recovery required several minutes before a repeat measurement could be performed on the same sensor. As expected, when 200 μM K_2HPO_4 was added as a negative control, there was no signal, indicating that no ion pumping had occurred. Replacing the substrate with IDP resulted in a signal about half that of K_4PP_i . However, in the presence of 50 μM ETD or 50 μM ZLD, the signals were barely detectable, indicating no Na^+ pumping was observed.

This observation is consistent with the TmPPase:ETD and TmPPase:ZLD structures, which show the absence of Na^+ density in the ion gate. Interestingly, in all the solved TmPPase structures, Na^+ at the ion gate has only been observed in the IDP-bound structures (**Fig. 6B-E** [↗](#) and **Fig. EV7** [↗](#)). In the IDP-bound structure, four key residues (D703^{16.46}, D243^{6.50}, S247^{6.54} and E246^{6.53}) in the ion gate constitute the Na^+ binding site (**Fig. 6E** [↗](#)). The formation of the site is driven by the downward motion of TMH16 (**Fig. EV7A** [↗](#)), transitioning from the resting state (TmPPase:Ca) to the closed state (TmPPase:IDP). The orientation of D703^{16.46} of the TmPPase:ETD structure resembles the structure of TmPPase:Ca, rotated away from the Na^+ binding site, causing a loss of Na^+ binding (**Fig. 6B,C** [↗](#)). In the TmPPase:ZLD structure, D703^{16.46} and K707^{16.50} are oriented relatively similarly to the Na^+ binding position in the TmPPase:IDP structure (**Figs. 6D,E** [↗](#) and **Fig. EV7C** [↗](#)), but no Na^+ density was observed despite the higher resolution compared to the TmPPase:IDP structure (3.26 Å compared to 3.5 Å for the IDP-bound structure). This might be because the inhibitor restricts the complete closure of the active site and full constriction and downward movement of the inner helices (especially TMH12 and 16) (**Fig. 3A-D** [↗](#)), which hinder the Na^+ pumping.

Figure 5.

Combined model of Ca (monomer A) and IDP (monomer B) structures

A. Side and bottom views of the symmetric and asymmetric structures for IDP bound to TmPPase. The asymmetric model combines a monomer (A) of TmPPase_Ca and a monomer (B) of TmPPase_IDP. **B.** The overlap of their predicted distance distribution and resulting experimental DEER distance distribution for S525R1.

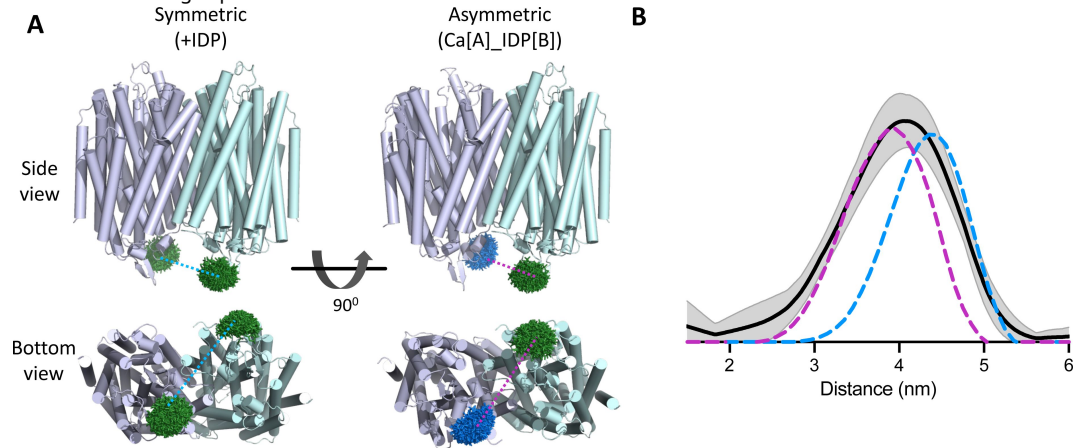
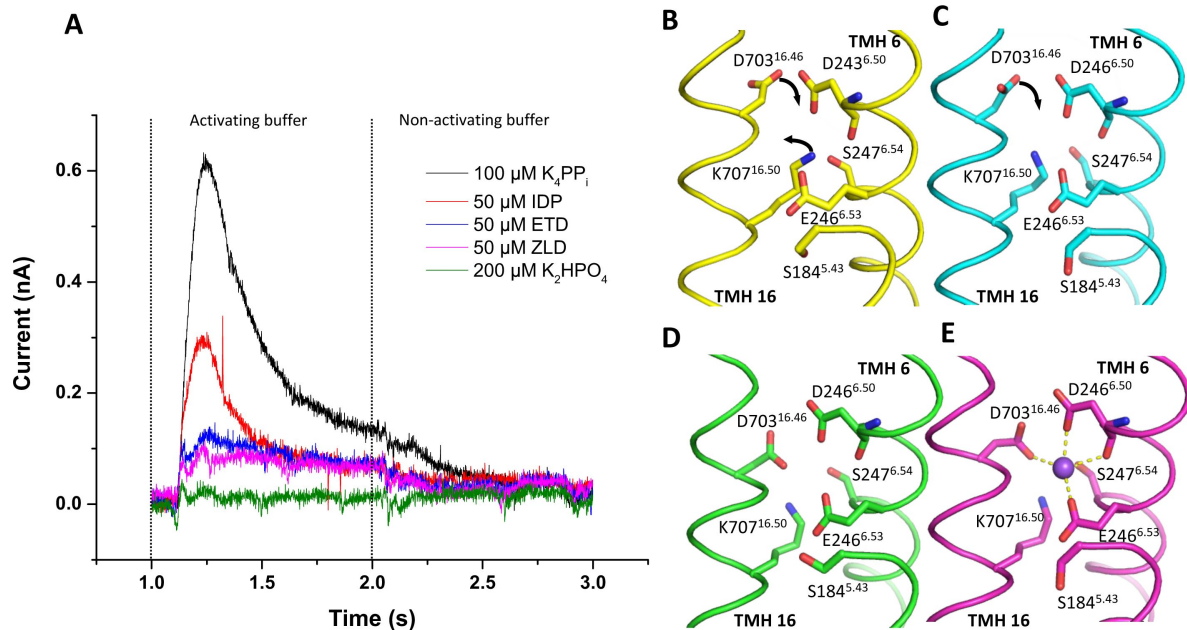


Figure 6.

Transient currents of TmPPase Na⁺ pumping and ion gate of TmPPase structures.

A. Curve of Na⁺ pumping current triggered by 100 μ M of K₄PP_i, 50 μ M of IDP, 50 μ M of ETD, 50 μ M of ZLD, and 200 μ M of K₂HPO₄. The vertical black dashed line represents the addition of activating buffer and non-activating buffer. **C-E.** Ion gate of TmPPase:Ca (yellow); TmPPase:ETD (cyan); TmPPase:ZLD (green); TmPPase:IDP (purple). The black arrows show the movement of residues of D703^{16.46} and K707^{16.50}.



Discussion

Inhibition of TmPPase by bisphosphonates

The seven distinct bisphosphonates we tested exhibit varying levels of inhibition against TmPPase (**Fig. 1B**). ETD and IBD exhibited higher IC_{50} values compared to PMD, ALD, and NRD ($p < 0.0001$) (**Fig. 1B**). This is consistent with the K_i of ETD on DhPPase (mPPase of *Desulfotobacterium hafniense*), which is approximately 67 times higher than that of amino methylene diphosphonate (AMDP), as measured by Viktor *et al.*^{22,53}. The difference may be due to the introduction of an amino group in the side chain and the length of the side chain⁴¹. Substituting the hydrogen (in ETD) with the benzene ring (in ZLD and RSD) does not favour inhibitory activity ($IC_{50} > 200 \mu M$). Nonetheless, these aromatic-containing compounds are still capable of preventing NEM modification on TmPPase (**Fig. 1C**), as further supported by the solved structure of ZLD bound with TmPPase (see discussion below).

Catalytic asymmetry in mPPase

Some evidence for asymmetry in mPPase gating has been shown previously by kinetic studies^{7,22} and captured in the time-resolved 600s and 3600s structures of TmPPase:PP_i⁷, where in both structures, one chain is in the open state and the other is in the closed state. Our new structures, together with DEER distance measurements that monitor the conformational ensemble equilibrium of TmPPase in solution, provide further solid experimental evidence of asymmetry in gating and transitional changes upon substrate/inhibitor binding.

In our study, the TmPPase:ETD structure captured the asymmetric binding of ETD (**Fig. 2**). Loop5-6, which interact with ETD, moves inward to close the active site, but not as deeply as in the TmPPase:IDP structure (**Fig. 2A,B**). Here, ETD is positioned²² above the hydrolytic centre (**Fig. 2B**), unable to descend further due to occupation of the Ca^{2+} in the active site (**Fig. 2C,D**), as in the TmPPase:Ca structure (PDB ID: 4AV3). This is supported by our DEER data recorded in solution, which showed an enhancement of the longer distance distribution components of T211R1 when comparing the +ETD-only to the +Ca/ETD condition (**Fig. EV8D**), indicating a semi-closed conformation. While in the absence of Ca^{2+} , the loops move closer together. Due to the different orientations of ETD in the two monomers, the orientations of loop5-6 differ as well (**Fig. 2A,B**). Nevertheless, the TmPPase:ETD structure can be considered as a snapshot transition between the *apo* structure (TmPPase:0-60s, PDB: 8B21), where no substrate is bound, and the TmPPase:600s structure (PDB: 8B23), where the PP_i has bound to one chain and descended to the hydrolytic centre. On the other hand, the TmPPase:ZLD structure, while resembling the IDP-bound structure, is more open (**Fig. 4**) and ZLD is positioned about 1.0 Å above the IDP position in the hydrolytic centre (**Fig. 3D**)⁷.

Based on these observations, and the DEER data for +IDP, which is consistent with an asymmetric conformation of TmPPase being present in solution, we propose five distinct models of TmPPase (**Fig. 7**). All these models involve one monomer in an open/resting state, while the other is constrained upon the binding of the substrate/inhibitors. Model 1 (**Fig. 7**), where both monomers are in the resting state as in the TmPPase:Ca structure⁸, displays both active sites in an open conformation. In model 2 (**Fig. 7**), one active site is semi-closed, while the other remains open. This is supported by the distance distributions for S525R1 and T211R1 for +Ca/ETD informed by DEER, which agrees with the *in silico* distance predictions generated by the asymmetric TmPPase:ETD X-ray structure.

While models 1 and 2 differ at the active site, the DEER distance distributions recorded for the S525R1 pair in the near exit channel are highly similar (**Fig. 7**, models 1 and 2). However, in the presence of ZLD, and as seen in model 3 (**Fig. 7**), the mean of the DEER distance distribution for

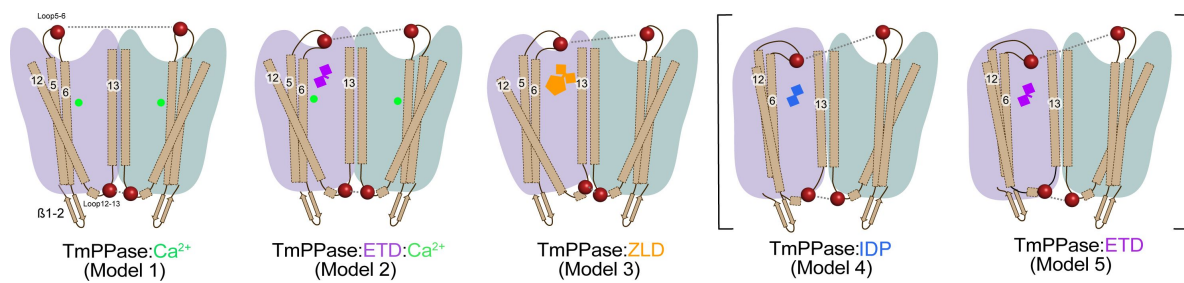


Figure 7.

Models based on DEER distance distributions for TmPPase S525R1 and T211R1.

Five DEER models of PfPPase bound bisphosphonates. TMH12, 13 and loop12-13 are shown in brown; TMH5, 6 and loop5-6 are shown in green. The labelling sites are represented by maroon spheres. Within the structures, Ca²⁺ is shown as a green circle; ETD by a purple square; ZLD by an orange pentagon and IDP as blue square.

S525R1 is shifted towards shorter distances (3.4 nm), while the active site (T211R1) retains the configuration observed in model 2. This might be because, although ZLD can enter the active site, its bulky heteroaryl group prevents the complete closure of the active site and restricts the full downward movement of TMH12 and TMH16. This likely prevents the formation of the Na⁺-binding site, leading to the absence of Na⁺ in the TmPPase:ZLD structure (Fig. 3) and the lack of Na⁺ signal in the electrometric measurement (Fig. 5).

Among all conditions tested, the DEER distance distributions of IDP-bound TmPPase exhibit the longest distances recorded for the periplasmic S525R1 site, but the shortest for the cytoplasmic T211R1 site. This suggests that the binding of IDP to TmPPase induces the closure of the active site and the full downward movement of TMH12 and TMH16, in line with the IDP-bound TmPPase structure. However, the DEER distance distributions obtained in solution for both S525R1 and T211R1 fall in between the extreme modelled distances seen in the X-ray structures. This is due to the presence of an asymmetric conformation between the two monomers, consistent with the asymmetric binding model (Fig. 7, model 4,5), where IDP (and similarly for ETD without Ca²⁺ present) binds only to one monomer, as in the hybrid structure TmPPase:Ca(A)_IDP(B) (Fig. 5A).

These models further confirm structural and functional studies showing the asymmetry in the catalysis of TmPPase. Together with our previous time-resolved structures⁷, they complement the ensemble of conformational changes during the reaction steps.

Sodium ion pumping in TmPPase

In a previous study⁷, we found that the single turnover event of Na⁺ pumping only occurs in the presence of IDP. In our current Nanion SURFET²R experiment, upon the addition of ETD and ZLD, we did not observe Na⁺ pumping (Fig. 5A), consistent with ETD and ZLD bound structures where no Na⁺ was observed at the ion gate.

We proposed that the motion of TMH12 and the communication network from TMH5 to TMH13 and TMH10 are key parts of intra-subunit communication between the two monomers^{7,20}. Loop12-13, where the S525R1 site is located, can be used to monitor the motion of TMH12. The mean distance for the S525R1 distribution in the *apo* state is shorter compared with the IDP bound state (Fig. 4D). However, in the presence of ZLD, the distances for S525R1 are even shorter than in *apo*, contradicting our initial prediction of having similar distances to the IDP bound state. This suggests that in solution, the exit channel of the ZLD-bound TmPPase can adopt a conformation that resembles the *apo* state, even though its active site is similar to the IDP-bound condition. This may explain why TmPPase is unable to pump Na⁺ upon the addition of ZLD. We suggest that while IDP constrain the conformation of one subunit to a closed state as in model 4 (Fig. 4D) and facilitate single turnover events of Na⁺ transport⁴⁵, ZLD and ETD restrict the conformation of one subunit to an intermediate state as in models 3 and 5 (Fig. 7), which is unable to facilitate Na⁺ transport.

In summary, EPR experiments in solution, coupled with new structures of inhibited forms of TmPPase, reveal that TmPPase binds inhibitors asymmetrically, forming open-closed states, where one monomer is in the inhibitor (IDP, ETD, ZLD) bound state and the other monomer is in the open state, consistent with half-of-the-sites-reactivity⁷. In future studies, we could use time-resolved DEER in complex with the substrate to explore, for example, how and in what order the release of phosphate products and ions leads to conformational changes in the active sites, exit channels, and communication between subunits, capturing further details during the reaction steps.

Materials and methods

Protein expression and purification

TmPPase expression and purification have been described previously^{54,55}. Briefly, pRS1024 plasmid containing his-tagged TmPPase was freshly transformed into *Saccharomyces cerevisiae* strain BJ1991. The cells were cultured in 250 ml of selective synthetic complete drop-out (SCD) media overnight before being added to 740 ml of 1.5× YP media with 2% glucose. The cells were then cultured for 8 h at 30 °C, collected by centrifugation (4,000 rpm, 10 min) and lysed at 4 °C using a bead beater with 0.2 mm glass beads. The membrane fraction was collected by ultracentrifugation (100,000 × g, 45 min) and the pellets were resuspended in buffer containing 50 mM MES-NaOH pH 6.5, 20% (v/v) glycerol, 50 mM KCl, 5.2 mM MgCl₂, 1.33 mM dithiothreitol (DTT), 2 µg ml⁻¹ (w/v) pepstatin-A (Sigma) and 0.334 mM PMSF (Sigma). The membranes were solubilised in solubilisation buffer (50 mM MES-NaOH pH 6.5, 20 % (v/v) glycerol, 5.33 % (w/v) n-dodecyl-β-D-maltopyranoside (DDM) (Anatrace)) using the ‘hot-solve’ method⁵⁵ at 75 °C for 1.5 h. After centrifugation to remove denatured proteins, KCl (to a final concentration of 0.3 M) and 2 ml of Ni-NTA beads (Qiagen) were added and incubated at 40 °C for 1.5 h, and then loaded into an Econo-Pac® column (Bio-Rad). Then the column was washed with two column volume (CV) of washing buffer (50 mM MES-NaOH pH 6.5, 20% (v/v) glycerol, 50 mM KCl, 20 mM imidazole pH 6.5, 5 mM MgCl₂, 1 mM DTT, 2 mg/ml (w/v) pepstatin-A, 0.2 mM PMSF and 0.05% DDM (Anatrace) and eluted with 2 CV of elution buffer (50 mM MES-NaOH pH 6.5, 3.5% (v/v) glycerol, 50 mM KCl, 400 mM imidazole pH 6.5, 5 mM MgCl₂, 1 mM DTT, 2 mg/ml (w/v) pepstatin-A, 0.2 mM PMSF and 0.5% octyl glucose neopentyl glycol (OGNPG, Anatrace).

TmPPase activity assay

TmPPase activity and bisphosphonates inhibition assay were performed using the molybdenum blue reaction method in 96-well plate format as reported previously⁴². Before the assay, the enzyme was reactivated by adding to the mixture of 30 mg/ml of soy-bean lecithin (Sigma) in 20 mM Tris-HCl pH 8.0 with 4.5% DDM and incubated at 55 °C for 15 min. The activity reaction was done in the reaction buffer (60 mM Tris-HCl pH 8.0, 5 mM MgCl₂, 100 mM KCl, and 10 mM NaCl) and started by adding 2 mM Na₄PP_i at 71 °C for 5 min.

NEM modification assay

The NEM modification assay was performed as reported previously^{8,56} with slight modification. Briefly, 0.4 mg/ml of the reactivated TmPPase was mixed with the modification buffer (20 mM MES-KOH pH 6.5, 0.05% DDM, 2.4 mM MgCl₂, 100 mM KCl, and 20 mM NaCl) and different inhibitors (2 mM CaCl₂, 0.5 mM IDP, and 0.5 mM of bisphosphonates) and incubated on ice for 30 min. Afterwards, 100 mM *N*-ethyl maleimide (NEM) (Thermo Scientific) was added and the mixture was further incubated for 10 min. The NEM-modification reactions were stopped by adding 2 mM DTT and the residual activity of the enzyme was performed using the molybdenum blue reaction assay⁵⁷.

Crystallisation and structure determination

For co-crystallisation with bisphosphonates, the purified TmPPase was buffer exchanged to the crystallisation buffer (50 mM MES-NaOH pH 6.5, 3.5% (v/v) glycerol, 50 mM KCl, 5 mM MgCl₂, 2 mM DTT and 0.5% OGNPG) on a Micro Bio-Spin 6 column (Bio-Rad) and then diluted to a concentration of 10 mg/ml. Prior to crystallisation, 1 mM bisphosphonates was added to the protein solution, incubated at room temperature for 30 min, and centrifuged for 20 min (16,000 g, 4 °C). Crystallisation trials were done using a Mosquito robot (SPT Labtech) by sitting drop vapour-diffusion method using MemGold™ screen (Molecular Dimensions) in MRC 2-well crystallisation plates (Swissci), and the drops were monitored at 22 °C using the minstrel DT UV imaging system (Formulatrix). Crystal hits appeared on the MemGold™ screen under different conditions. Harvestable crystals appeared within several days and were frozen directly from the mother liquor. For the TmPPase cocrystallised with etidronate, the best diffracting crystal was observed

from a solution containing 0.2 M CaCl_2 , 0.1 M HEPES pH 7.0, and 33% PEG400, while for TmPPase cocrystallised with zoledronate, the best diffracting crystal was observed from a solution containing 0.1 M MES pH 6.5, 0.1 M NaCl, 33% PEG400, and 4% ethylene glycol.

X-ray diffraction data were collected at Diamond Light Source (DLS) (UK) on the I03 (TmPPase:ETD) and I04-1 beamline (TmPPase:ZLD) at 100 K on a Pilatus 6M detector. The data were merged and scaled using X-ray Detector Software (XDS)⁵⁸ and the structure was solved by molecular replacement with Phaser⁵⁹ using the resting state (4AV3)^{8,56} and IDP-bound (5LZQ) state⁹ of TmPPase structure as the search model for TmPPase:Etidronate and TmPPase:Zoledronate, respectively. The structures were built and refined using phenix.refine⁶⁰ and Coot⁶¹. X-ray data and refinement statistics are listed in Table 1.

EPR Spectroscopy

Sample preparation for EPR spectroscopy

For EPR spectroscopy measurements, residue S525, located in the periplasmic loop12-13 of the TmPPase exit channel, was mutated to cysteine and covalently modified with a methanethiosulfonate thiol-specific spin label (MTSSL) to introduce a paramagnetic centre^{62,63} (the labelled protein is referred to as S525R1). At the cytoplasmic side of the membrane interface, we constructed the TmPPase T211C variant, which is located in loop5-6 and above the active site (the MTSSL labelled mutant is referred to as T211R1).

The S525C and T211C proteins were expressed as outlined above. The frozen cell pellets were lysed using cryo-milling (Retsch model MM400). 1 mM TCEP was used to replace DTT in the purification steps preceding spin labelling and the remaining purification was carried out as above. Each protein was spin-labelled with MTSSL while immobilised to the Ni-NTA resin (or mixed following Cys mutant elution) as previously described^{39,48}. Briefly, for MTSSL labelling, MTSSL was added in spin-label buffer (20 mM MOPS-NaOH, 1 mM TCEP, 5 mM MgCl_2 , 50 mM KCl, 3.5% glycerol, 0.03% DDM at pH 7.5) at 10-fold molar protein excess and incubated for 2 hours at room temperature. Spin-labelled protein was eluted from the Ni-NTA resin column, concentrated and subsequently purified using size-exclusion chromatography using Superose 6 increase 10/300 GL (GE Healthcare) and equilibrated in 20 mM MES-NaOH, pH 6.5, 5 mM MgCl_2 , 50 mM KCl, 3.5% glycerol, 0.05% DDM. The eluted purified protein fractions were concentrated, buffer exchanged with buffer prepared in D_2O , and split into aliquots for incubation with a final concentration of 2 mM of inhibitors or 10 mM CaCl_2 (30 min, RT). The protein activity was tested as described above, and the protein samples were tested for spin labelling by CW EPR spectroscopy, and then 40 % ethylene glycol- d_6 was added to each sample before flash freezing for DEER measurement.

Continuous Wave EPR (CW EPR) spectroscopy

CW EPR experiments were performed on a Bruker Magnetech ESR5000 X-band spectrometer. The spin-labelled sample was loaded into an EPR tube before the addition of ethylene glycol- d_6 . The samples were measured at room temperature, as TmPPase is more thermally stable than most membrane proteins. The measurements were performed in a 330-345 mT magnetic field, with a 60 s sweep time, 0.1 mT modulation, 100 kHz frequency, and 10 mW (10 dB) microwave power.

Double Electron-Electron Resonance (DEER, or PELDOR) spectroscopy

DEER distance measurements and set-up

EPR recordings were collected as previously described⁶⁴ using a Bruker ELEXSYS E580 Spectrometer operating at Q-band frequency, equipped with a QT-II resonator in a Cryogenic Ltd. cryogen-free variable temperature cryostat for EPR. In brief, spin-labelled protein samples were prepared in 3 mm outer diameter quartz tubes and data was recorded at 50 K. The detection pulse sequence used was a refocused Hahn echo: $\pi/2 - \tau_1 - \pi - \tau_1 - \tau_2 - \pi - \tau_2 - \text{echo}$, with $\pi/2$ and π pulse lengths of 32 ns, τ_1 of 380 ns and τ_2 of 2000 ns. In all cases, the frequency of the detection pulses was 33.96 GHz, and a 16 ns pump pulse of 34.025 GHz was moved in 4 ns increments in the gap between the two π pulses of the detection sequence to yield the DEER trace. All pulses were generated using an integrated Bruker SpinJet AWG, and a 16-step phase cycle on the detection pulses was used to remove unwanted echo crossings⁶⁵. Traces were recorded at a magnetic field of 12111 G, setting the pump pulse resonant with the nitroxide spectral maximum. Scans were recorded until a sufficient signal-to-noise ratio was obtained, typically with datasets averaged overnight.

DEER data analysis and processing

Distance distributions were determined from the time traces using various methodologies as this becomes a good practice to get reliable results and to show consistency between different approaches. In the present work, we used two different programs, DeerAnalysis^{66,67} and ComparativeDeerAnalyzer⁶⁷ which are MATLAB-based programs. Data were processed using DeerAnalysis2022. The data were loaded, the “2+1” artefact truncated from the dataset (-600 ns)⁶⁸, then phase and background corrected using the ‘!’ automated adjustment. The background corrected traces were then transformed from the time domain to the distance domain using Tikhonov Regularization⁶⁹, and the quality of the fit was assessed based on the L-curve method and the shape of the Pake pattern. The resultant background correction was then validated using a module for validation implemented in DA. The validation was carried out after initial Tikhonov regularisation, varying the background start time from 5% to 80% of the respective time windows of the cut data for 16 trials. From this, the raw data were re-loaded and processed (Tikhonov regularisation) with the cutoff and background start time as established from the first round of validation. Resulting data are taken for plotting figures showing raw data/background and form factor/fit. This is the starting point for a full validation, where the background start time was again varied from 5% to 80% of the time window for 16 trials, as well as some added “white noise” with a level of 1.50 for 50 trials. The result of validation was pruned and used for plotting the distance distribution and confidence interval. We have also used the ComparativeDeerAnalyser (CDA) feature implemented in DeerAnalysis 2022 which is an automated process that uses three approaches simultaneously to assess and reduce model bias by comparing their associated results.

In silico spin labelling and modelling

MtsslWizard⁵⁰ was used to analyse the labelling sites of TmPPase to predict the shifts in distance distribution under different conditions. The PDB coordinates of the respective X-ray structures (TmPPase:Ca, TmPPase:Ca:ETD, TmPPase:ZLD, and TmPPase:IDP) for the different conditions were uploaded to the online MTSSL Suite server, labelled at T211 and S525 sites (both monomer A and B) using the mode ‘tight 0 clashes allowed’. The distance distributions were then calculated. The distance distribution data were exported to Prism 10 (GraphPad) to plot the distance distribution graphs.

Electrometric measurement

For the Nanion SURFE²R experiment, purified TmPPase was reconstituted into liposomes as previously described⁴⁵ with some modifications. Briefly, the purified protein was buffer exchanged into a reconstitution buffer (50 mM MOPS-KOH pH 7.2, 50mM KCl, 5mM MgCl₂, and 2mM DTT) to remove Na⁺ and glycerol and then diluted to 50 µg/ml concentration. 15 µl of liposome solution (120 mg/ml soy-bean lecithin in 50 mM MOPS-KOH pH 7.2) was mixed with 1 ml of diluted protein sample. SM-2 Bio-beads were added in increments to a final concentration of 0.25 mg/µl and then placed into a mixer at 4 °C for 6 h to ensure beads stayed in suspension. The proteoliposomes were collected and frozen at -80 °C in aliquots. To ensure that the reconstituted protein was still active, the hydrolytic activity was performed using the molybdenum blue reaction assay⁵⁷.

Electrometric measurements were performed on a SURFE²R N1 instrument from Nanion Technology (Munich, Germany). The gold sensors were prepared based on the ‘SURFE²R N1 protocol’, including thiolation of the sensor surface and assembly of the lipid layer using sensor prep A2 and B solutions. 15 µl of sonicated proteoliposomes, followed by 50 µl of the rinsing buffer (50 mM MOPS-KOH pH 7.2, 50 mM NaCl, 5 mM MgCl₂) were applied directly to the sensor surface. Sensors were centrifuged for 30 minutes at 2500 g and incubated at 4 °C for 3 h. The sensors were mounted in the SURFE²R N1 and rinsed once with 1 ml of rinsing buffer (50 mM MOPS-KOH, pH 7.2, 50 mM NaCl, 5 mM MgCl₂). Measurements were performed for 3 s by consecutively flowing non-activating buffer B (50 mM MOPS-KOH pH 7.2, 50 mM NaCl, 5 mM MgCl₂, 200 µM K₂HPO₄) and activating buffer A (50 mM MOPS-KOH, 50 mM NaCl, 5 mM MgCl₂) containing substrate (100 µM K₄PP_i) or inhibitors (50 µM IDP, 50 µM ETD or 50 µM ZLD) across the sensor for 1 s each in a BAB sequence. Charge transport across the membrane is initiated by substrate or inhibitor in buffer A, which flows across the sensor between 1 and 2 s. The transport of positively charged ions during this period results in a positive electrical current, the signal output of the SURFE²R N1 instrument. Between each measurement, the sensor was washed with 1 ml rinsing buffer and incubated for 60 seconds. The measurements were tested in triplicates.

Data availability

The atomic coordinates and structure factors of the TmPPase:Etidronate and TmPPase:Zoledronate complex have been deposited in the Protein Data Bank, www.rcsb.org (PDB ID: 9G8K and 9G8J).

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Additional information

Author contributions

Conceptualisation, JL, KV, CP, HX, AG; Methodology, JL, KV, AS, KH, ABr, ABo, AG, CP, NGJ, LJ, Investigation, KV, JL, AS, YM, OR; Resources, AG, CP, KV, NGJ, HX, JYK.; Writing – Original draft, JL, AS, KV, CP; Writing-Review and Editing, JL, KV, NGJ, CP, AS, LJ, JYK, HX, AG; Visualization, KV, CP, AS, JL; Supervision and Project Administration, AG, CP, HX, JYK.; Funding Acquisition, JL, KV, CP, AG, HX, JYK.

Declaration of interests

All authors declare no conflict of interest in this paper.

Supplementary figures and tables

Expanded View figures

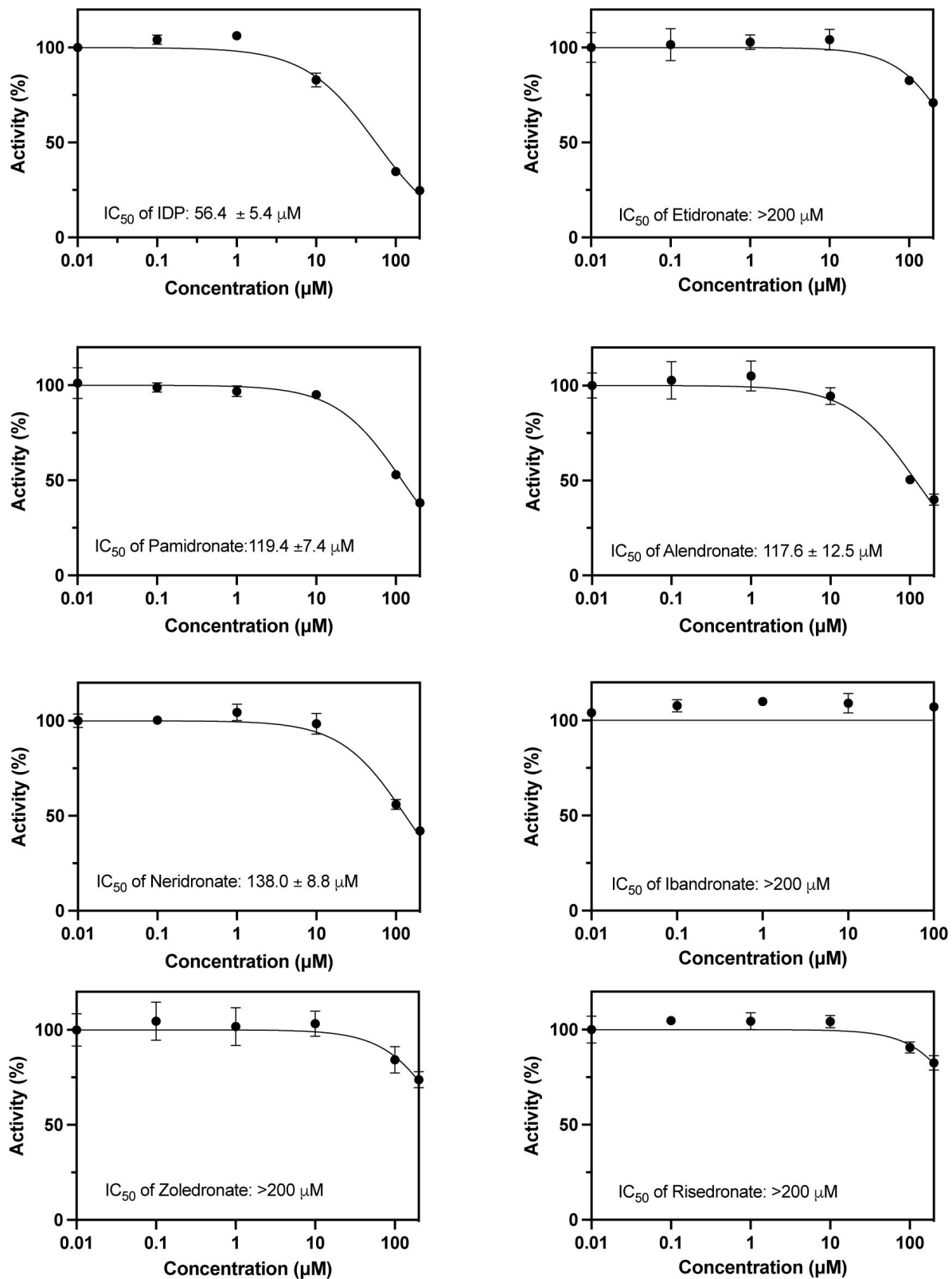


Figure EV1.

Inhibition of TmPPase by bisphosphonates.

All data are shown as mean $\pm$ SD with three replicates.

Figure EV2.

Accessible cysteines for NEM modification.

A. Two exposed cysteines in both monomers of TmPPase:Ca (PDB: 4AV3; cyan). **B.** One exposed cysteine in both monomers of TmPPase:IDP (PDB: 5LZQ; wheat).

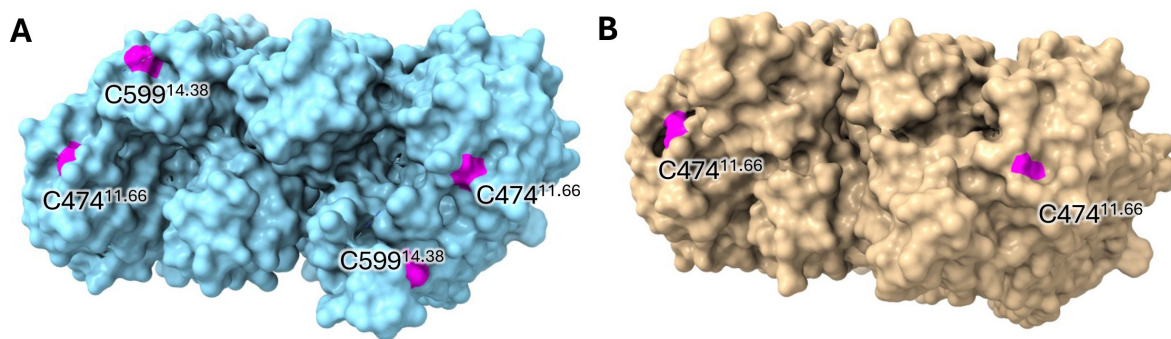


Figure EV3.

Electron density maps of ETD at the active sites.

A. mF_o-F_c omit map with positive density of the ETD_A shown in green mesh (*Top left*); Polder omit map of ETD_A (*Bottom left*) shown in yellow mesh; 2mF_o-F_c (blue mesh) map of ETD_A, ions and surrounding residues (*Right*). **B.** mF_o-F_c omit map with positive density of the ETD_B shown in green mesh (*Top left*); Polder omit map of ETD_B (*Bottom left*) shown in yellow mesh; 2mF_o-F_c (blue mesh) map of ETD_B, ions and surrounding residues (*Right*). Ca²⁺ ion is shown in purple; Mg²⁺ ions are shown in green and water molecules are shown in red.

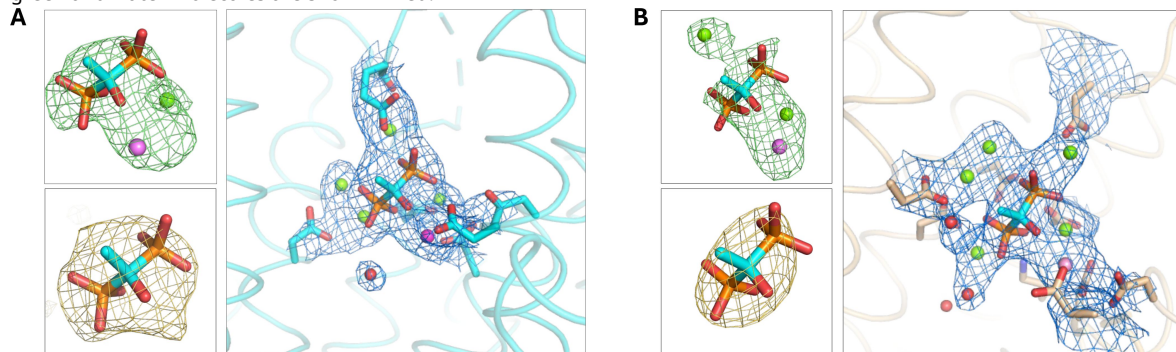


Figure EV4.

Electron density maps of ZLD at the active sites.

A. $mF_o - F_c$ omit map with positive density of the ZLD_A shown in green mesh (*Top left*); Polder omit map of ZLD_A (*Bottom left*) shown in yellow mesh; $2mF_o - F_c$ (blue mesh) map of ZLD_A , ions and surrounding residues (*Right*). **B.** $mF_o - F_c$ omit map with positive density of the ZLD_B shown in green mesh (*Top left*); Polder omit map of ZLD_B (*Bottom left*) shown in yellow mesh; $2mF_o - F_c$ (blue mesh) map of ZLD_B , ions and surrounding residues (*Right*). **C.** $mF_o - F_c$ omit map with positive density of the ZLD_C shown in green mesh (*Top left*); Polder omit map of ZLD_C (*Bottom left*) shown in yellow mesh; $2mF_o - F_c$ (blue mesh) map of ZLD_C , ions and surrounding residues (*Right*). **D.** $mF_o - F_c$ omit map with positive density of the ZLD_D shown in green mesh (*Top left*); Polder omit map of ZLD_D (*Bottom left*) shown in yellow mesh; $2mF_o - F_c$ (blue mesh) map of ZLD_D , ions and surrounding residues (*Right*).

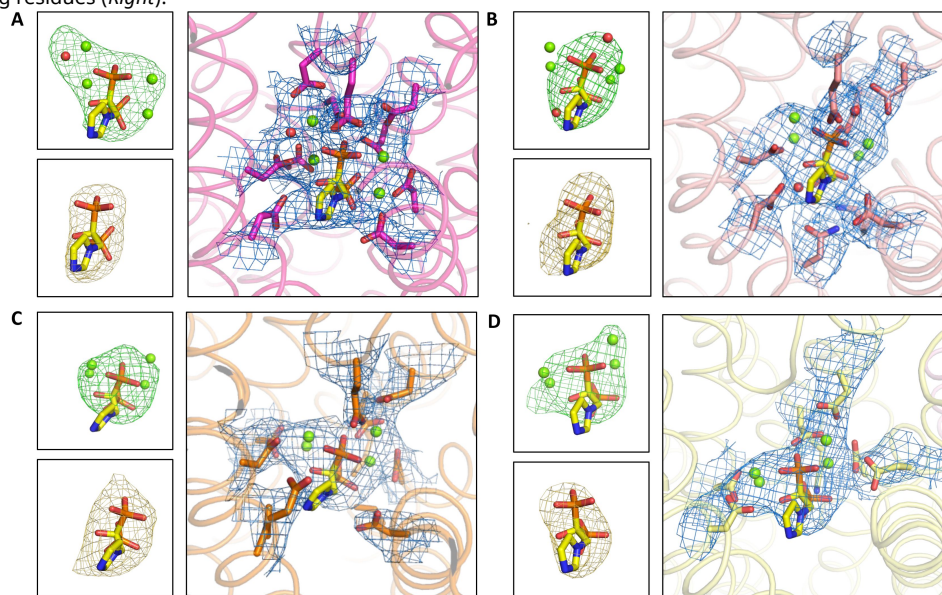


Figure EV5.

Electron density maps of loop5-6 in the TmPPase:ETD structure.

A. $2mF_o - F_c$ (blue mesh) electron density map of loop5-6 at chain A. **B.** $2mF_o - F_c$ (blue mesh) electron density map of loop5-6 at chain B.

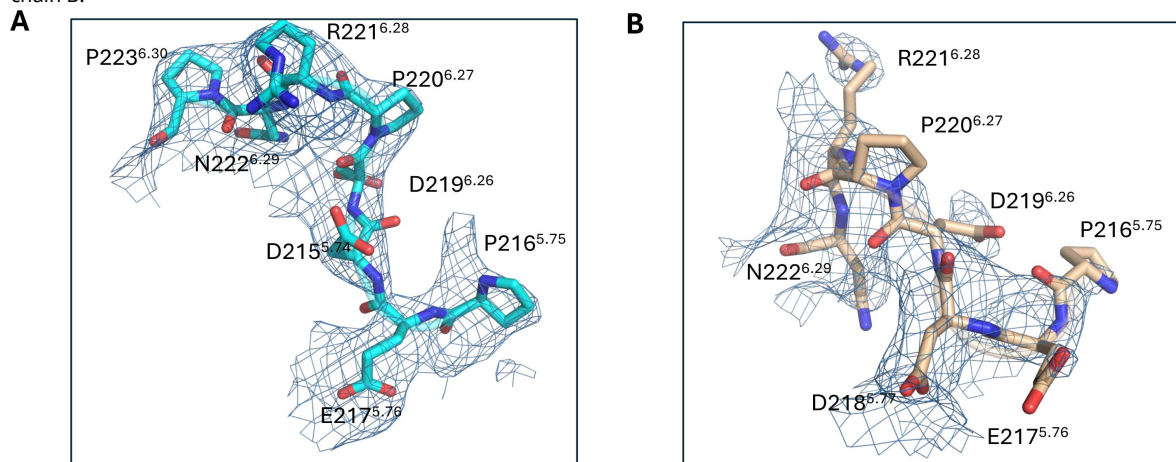


Figure EV6.

Helix curvature comparison between chain A and chain B of the TmPPase:ETD structure.

Changes in helix curvature are shown in the hydrolytic side of TMH11, TMH12, and TMH15. The black bar shows the region in the hydrolytic centre side.

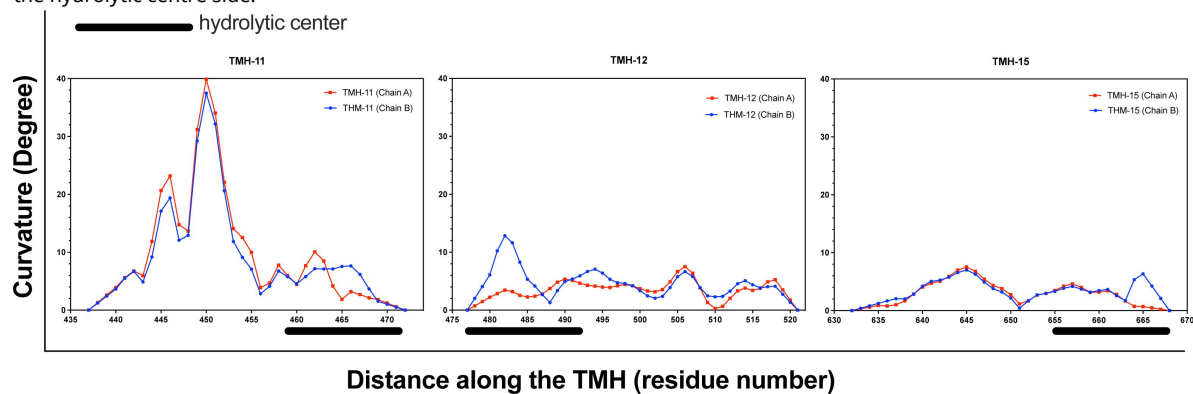
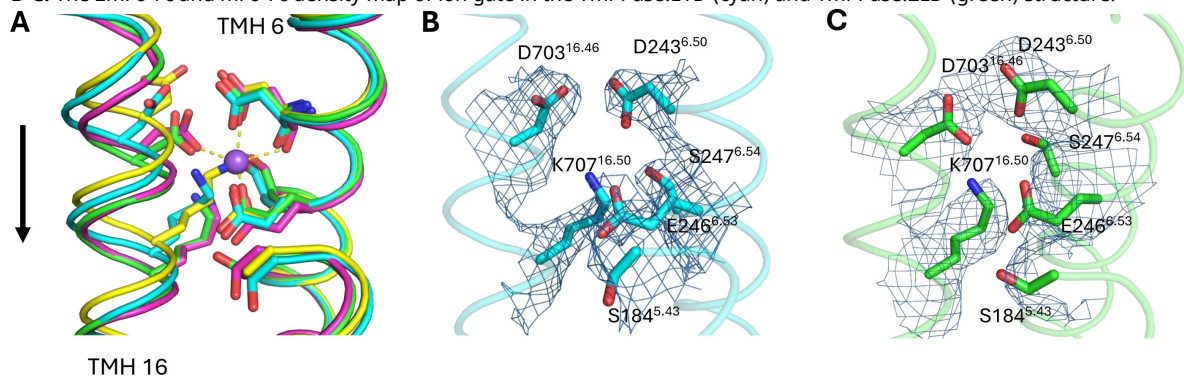


Figure EV7.

Ion gates of TmPPase structures.

A. Superposition of the ion gate of the four TmPPase structures (yellow: TmPPase:Ca; cyan: TmPPase:ETD; green: TmPPase:ZLD; purple: TmPPase:IDP; Na^+ is shown as a purple sphere). The movement of TMH16 is shown as the black arrow.

B-C. The 2mFo-Fc and mFo-Fc density map of ion gate in the TmPPase:ETD (cyan) and TmPPase:ZLD (green) structure.



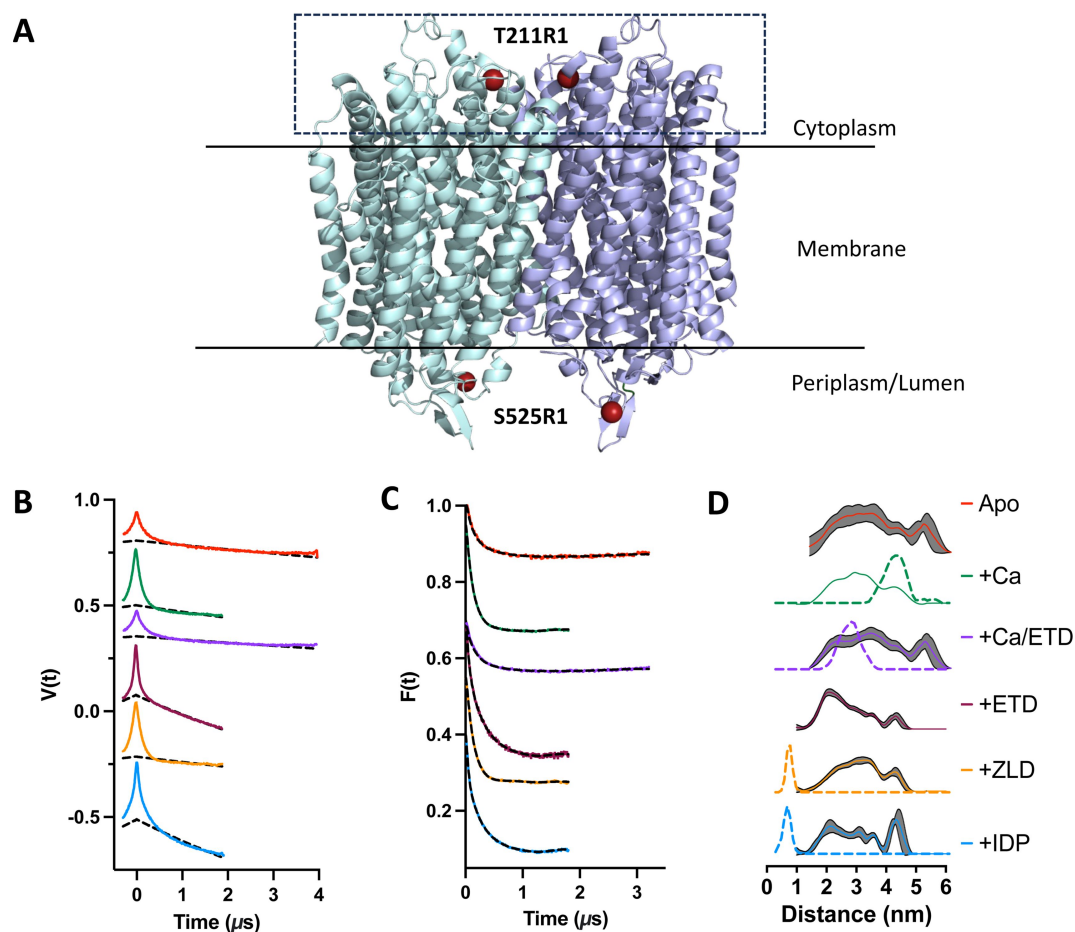


Figure EV8.

DEER distance distributions of TmPPase T211R1 under different conditions.

A. Structure of the TmPPase dimer (PDB: 5LZQ), with monomers A and B coloured cyan and purple, respectively. The sites that were mutated to cysteine and labelled with MTSSL are shown by maroon spheres, with T211R1 on the cytoplasmic (top) side and S525R1 on the periplasmic (bottom) side of the membrane interface. **B.** DEER raw data traces for T211R1. Each condition measured is coloured according to the condition used. **C.** DEER background-corrected time-domain traces for T211R1. **D.** The overlap between the predicted distance distribution of T211R1 from the solved crystal structures (TmPPase:Ca, TmPPase:Ca:ETD, TmPPase:ZLD, and TmPPase:IDP), shown as dashed lines, with the resulting DEER distance distributions at the respective conditions. The grey-shaded regions represent the uncertainty in the distribution. The data were all processed in DeerAnalysis, with validation in the same way as described in the methods.

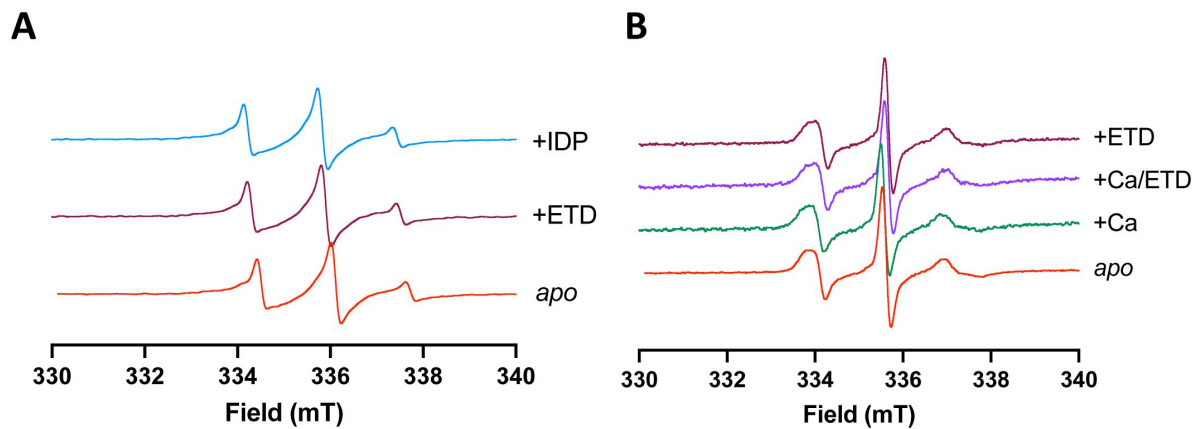


Figure EV9.

CW EPR spectra of TmPPase T211R1 and S525R1 under different conditions.

The CW EPR data were collected at room temperature before the addition of ethylene glycol and therefore represent the rotational motion of the spin labels in comparison to each other. **A.** The normalized CW EPR data for T211R1 in its *apo* form, and with ETD and IDP added. **B.** The normalized CW EPR data for S525R1 for *apo*, +Ca(II), +Ca/ETD and +ETD conditions.

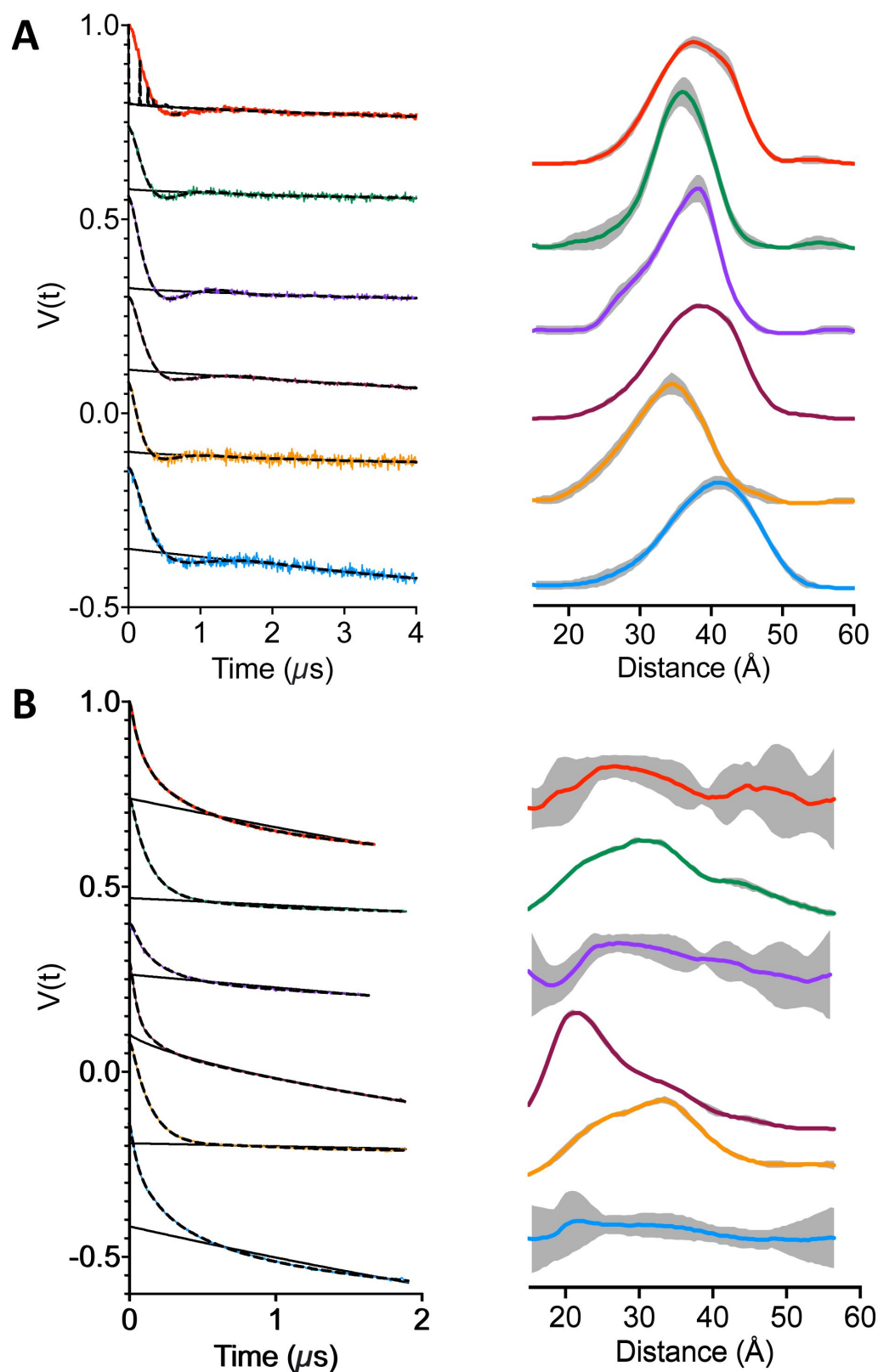


Figure EV10.

ComparativeDeerAnalyzer (CDA) data of TmPPase S525R1 and T211R1.

A. The background-corrected data and distance distributions for S525R1 generated by the automated CDA software. **B.** The background-corrected data and distance distributions for T211R1 generated by the automated CDA software.

Supplementary tables

Data Parameters	TmPPase+Etidronate	TmPPase+Zoledronate
Crystallisation condition	0.2M CaCl ₂ , 0.1M HEPES pH 7.0, 33% PEG 400	0.1M NaCl, 0.1M MES pH 6.5, 33% PEG 400, 4% ethylene glycol
Space group	P 2 ₁	P 2 ₁ 2 ₁ 2 ₁
Cell dimensions		
a, b, c (Å)	83.7, 111.7, 105.2	101.188, 147.366, 252.341
α, β, γ (°)	90.0, 106.7, 90.0	90.0, 90.0, 90.0
Source	DLS I03	DLS I04-1
Wavelength (Å)	0.91587	0.97625
Resolution (Å)	80.2 - 3.15 (3.56 - 3.15)	127.3 - 3.26 (3.31 - 3.26)
Overall (Å)	3.15	3.26
along h axis	3.10	4.46
along k axis	3.60	4.17
along l axis	4.31	3.17
Measured reflections	130818	445655
Unique reflections	19273	33220
Completeness (%)	91.6 (59.3)	93.7 (72.2)
CC _{1/2}	0.999 (0.494)	0.999 (0.573)
Mean I/ σ (I)	9.7 (1.6)	10.2 (1.7)
Multiplicity	6.8 (7.0)	13.4 (12.2)
Wilson B (Å ²)	98.7	118.03
R _{merge}	0.098 (1.193)	0.137 (1.785)
R _{meas}	0.106 (1.29)	0.143 (1.860)
R _{pim}	0.041 (0.486)	0.039 (0.516)

Table S1.

X-ray data collection and refinement statistics.

Refinement		
Resolution (Å)	74.81 - 3.15 (3.27 - 3.15)	48.22 - 3.27 (3.39 - 3.27)
R _{work} (%)/R _{free} (%)	27.2/31.0	25.9/30.4
No. of atoms	10043	20790
protein	10003	20704
ligands	44	104
water	8	6
No. of chains (ASU)	2	4
B-factors (Å ²)	88.96	106.62
Protein	88.85	106.53
Ligands/Ion	127.07	133.77
R.m.s. deviations		
Bond lengths (Å)	0.003	0.005
Bond angle (°)	0.63	0.76
Ramachandran statistics [†]		
Favoured (%)	97.73	98.59
Allowed (%)	2.27	1.41
Outliers (%)	0.00	0.00

Table S1. (continued)

Chain	ETD _A :ETD _B	4AV3 _A :4AV3 _B	ETD _A :4AV3 _A	ETD _B :4AV3 _B	ETD _A :4AV3 _B	ETD _B :4AV3 _A
C _α RMSD (Å)	1.44	0.39	0.72	0.94	0.70	0.98
Chain	ZLD _A :ZLD _B	5LZQ _A :5LZQ _B	ZLD _A :5LZQ _A	ZLD _B :5LZQ _B	ZLD _A : 5LZQ _B	ZLD _B :5LZQ _A
C _α RMSD (Å)	0.51	0.21	0.72	0.72	0.74	0.69
Chain	ZLD _A :ZLD _C	ZLD _A :ZLD _D	ZLD _C :ZLD _D	ZLD _B :ZLD _D	ZLD _B :ZLD _C	
C _α RMSD (Å)	0.58	0.68	0.90	0.61	0.66	

Table S2.
Structural alignments between chains of different TmPPase structures

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Reviewer #1 (Public review):

Summary:

This work examines the binding of several phosphonate compounds to a membrane-bound pyrophosphatase using several different approaches, including crystallography, electron paramagnetic resonance spectroscopy, and functional measurements of ion pumping and pyrophosphatase activity. The work attempts to synthesize these different approaches into a model of inhibition by phosphonates in which the two subunits of the functional dimer interact differently with the phosphonate.

Strengths:

This study integrates a variety of approaches, including structural biology, spectroscopic measurements of protein dynamics, and functional measurements. Overall, data analysis was thoughtful, with careful analysis of the substrate binding sites (for example calculation of POLDOR omit maps).

Weaknesses:

Unfortunately, the protein did not crystallize with the more potent phosphonate inhibitors. Instead, structures were solved with two compounds with weak inhibitory constants >200

micromolar, which limits the molecular insight into compounds that could possibly be developed into small molecule inhibitors. Likewise, the authors choose to focus the spectroscopy experiments on these weaker binders, missing an opportunity to provide insight into the interaction between more potent binders and the protein.

In general, the manuscript falls short of providing any major new insight into membrane-bound pyrophosphatases, which are a very well-studied system. Subtle changes in the structures and ensemble distance distributions suggest that the molecular conformations might change a little bit under different conditions, but this isn't a very surprising outcome. It's not clear whether these changes are functionally important, or just part of the normal experimental/protein ensemble variation.

The ZLD-bound crystal structure doesn't predict the DEER distances, and the conformation of Na⁺ binding site sidechains in the ZLD structure doesn't predict whether sodium currents occur. This might suggest that the ZLD structure captures a conformation that does not recapitulate what is happening in solution/ a membrane.

<https://doi.org/10.7554/eLife.102288.1.sa3>

Reviewer #2 (Public review):

Summary:

Crystallographic analysis revealed the asymmetric conformation of the dimer in the inhibitor-bound state. Based on this result, which is consistent with previous time-resolved analysis, authors verified the dynamics and distance between spin introduced label by DEER spectroscopy in solution and predicted possible patterns of asymmetric dimer.

Strengths:

Crystal structures with inhibitor bound provide detailed coordination in the binding pocket thus useful information for the PPase field and maybe for drug development.

Weaknesses:

The distance information measured by DEER is advantageous for verifying the dynamics and structure of membrane protein in solution. However, regarding T211 data, which, as the authors themselves stated, lacks measurement precision, it is unclear for readers how confident one can judge the conclusion leading from these data for the cytoplasmic side.

The distance information for the luminal site, which the authors claim is more accurate, does not indicate either the possibility or the basis for why it is the ensemble of two components and not simply a structure with a shorter distance than the crystal structure.

<https://doi.org/10.7554/eLife.102288.1.sa2>

Reviewer #3 (Public review):

Summary:

Membrane-bound pyrophosphatases (mPPases) are homodimeric proteins that hydrolyze pyrophosphate and pump H⁺/Na⁺ across membranes. They are attractive drug targets against protist pathogens. Non-hydrolysable PPi analogue bisphosphonates such as risedronate (RSD) and pamidronate (PMD) serve as primary drugs currently used. Bisphosphonates have a P-C-P bond, with its central carbon can accommodate up to two substituents, allowing a large compound variability. Here the authors solved two TmPPase structures in complex with the

bisphosphonates etidronate (ETD) and zoledronate (ZLD) and monitored their conformational ensemble using DEER spectroscopy in solution. These results reveal the inhibition mechanism of these compounds, which is crucial for developing future small molecule inhibitors.

Strengths:

The authors show that seven different bisphosphonates can inhibit TmPPase with IC₅₀ values in the micromolar range. Branched aliphatic and aromatic modifications showed weaker inhibition.

High-resolution structures for TmPPase with ETD (3.2 Å) and ZLD (3.3 Å) are determined. These structures reveal the binding mode and shed light on the inhibition mechanism. The nature of modification on the bisphosphonate alters the conformation of the binding pocket.

The conformational heterogeneity is further investigated using DEER spectroscopy under several conditions.

Weaknesses:

The authors observed asymmetry in the TmPPase-ELD structure above the hydrolytic center. The structural asymmetry arises due to differences in the orientation of ETD within each monomer at the active site. As a result, loop5-6 of the two monomers is oriented differently, resulting in the observed asymmetry. The authors attempt to further establish this asymmetry using DEER spectroscopy experiments. However, the (over)interpretation of these data leads to more confusion than any further understanding. DEER data suggest that the asymmetry observed in the TmPPase-ELD structure in this region might be funneled from the broad conformational space under the crystallization conditions.

DEER data for position T211R1 at the enzyme entrance reveal a highly flexible conformation of loop5-6 (and do not provide any direct evidence for asymmetry, Figure EV8). Similarly, data for position S521R1 near the exit channel do not directly support the proposed asymmetry for ETD. Despite the high quality of the data, they reveal a very similar distance distribution. The reported changes in distances are very small (± 0.3 nm), which can be accommodated by a change of spin label rotamer distribution alone. Further, these spin labels are located on a flexible loop, thereby making it difficult to directly relate any distance changes to the global conformation.

The interpretations listed below are not supported by the data presented:

(1) 'In the presence of Ca²⁺, the distance distribution shifts towards shorter distances, suggesting that the two monomers come closer at the periplasmic side, and consistent with the predicted distances derived from the TmPPase:Ca structure.'

Problem: This is a far-stretched interpretation of a tiny change, which is not reliable for the reasons described in the paragraph above.

(2) 'Based on the DEER data on the IDP-bound TmPPase, we observed significant deviations between the experimental and the in silico distances derived from the TmPPase:IDP X-ray structure for both cytoplasmic- (T211R1) and periplasmic-end (S525R1) sites (Figure 4D and Figure EV8D). This deviation could be explained by the dimer adopting an asymmetric conformation under the physiological conditions used for DEER, with one monomer in a closed state and the other in an open state.'

Problem: The authors are trying to establish asymmetry using the DEER data. Unfortunately, no significant difference is observed (between simulation and experiment) for position 525 as the authors claim (Figure 4D bottom panel). The observed difference for position 112 must be accounted for by the flexibility and the data provide no direct evidence for any asymmetry.

(3) 'Our new structures, together with DEER distance measurements that monitor the conformational ensemble equilibrium of TmPPase in solution, provide further solid experimental evidence of asymmetry in gating and transitional changes upon substrate/inhibitor binding.'

Problem: See above. The DEER data do not support any asymmetry.

(4) Based on these observations, and the DEER data for +IDP, which is consistent with an asymmetric conformation of TmPPase being present in solution, we propose five distinct models of TmPPase (Figure 7).

Problem: Again, the DEER data do not support any asymmetry and the authors may revisit the proposed models.

(5) 'In model 2 (Figure 7), one active site is semi-closed, while the other remains open. This is supported by the distance distributions for S525R1 and T211R1 for +Ca/ETD informed by DEER, which agrees with the in silico distance predictions generated by the asymmetric TmPPase:ETD X-ray structure'

Problem: Neither convincing nor supported by the data

<https://doi.org/10.7554/eLife.102288.1.sa1>

Author response:

Public Reviews:

Reviewer #1 (Public review):

Summary:

This work examines the binding of several phosphonate compounds to a membrane-bound pyrophosphatase using several different approaches, including crystallography, electron paramagnetic resonance spectroscopy, and functional measurements of ion pumping and pyrophosphatase activity. The work attempts to synthesize these different approaches into a model of inhibition by phosphonates in which the two subunits of the functional dimer interact differently with the phosphonate.

Strengths:

This study integrates a variety of approaches, including structural biology, spectroscopic measurements of protein dynamics, and functional measurements. Overall, data analysis was thoughtful, with careful analysis of the substrate binding sites (for example calculation of POLDOR omit maps).

Weaknesses:

Unfortunately, the protein did not crystallize with the more potent phosphonate inhibitors. Instead, structures were solved with two compounds with weak inhibitory constants >200 micromolar, which limits the molecular insight into compounds that could possibly be developed into small molecule inhibitors. Likewise, the authors choose to focus the spectroscopy experiments on these weaker binders, missing an opportunity to provide insight into the interaction between more potent binders and the protein.

We acknowledge the reviewer concern regarding the choice of weaker inhibitors. We attempted co-crystallization with all available inhibitors, including those with higher potency.

However, despite numerous efforts, these potent inhibitors yielded low-resolution crystals, making them unsuitable for detailed structural analysis. Therefore, we chose to focus on the weaker binders, as we were able to obtain high-quality crystal structures for these compounds. This allowed us to perform DEER spectroscopy with the added advantage of accurately analyzing the data against structural models derived from X-ray crystallography. Using these weaker inhibitors enabled a more precise interpretation of the DEER data, thus providing reliable insights into the conformational dynamics and inhibition mechanism. However, as suggested by the reviewer, in the revised version, we will perform DEER analysis on the more potent inhibitors to provide additional insight into their interactions.

In general, the manuscript falls short of providing any major new insight into membrane-bound pyrophosphatases, which are a very well-studied system. Subtle changes in the structures and ensemble distance distributions suggest that the molecular conformations might change a little bit under different conditions, but this isn't a very surprising outcome. It's not clear whether these changes are functionally important, or just part of the normal experimental/protein ensemble variation.

We respectfully disagree with the reviewer. The scale of motions seen in this study correspond to those seen in the full panoply of crystal structures of mPPases. Some proteins undergo very large conformational changes during catalysis – such as the rotary ATPase. This one doesn't, meaning that the precise motions we describe are likely to be relevant. Conformational changes in the ensemble, whether large or small, represent essential protein motions which underlie key mPPase catalytic function. Our DEER spectroscopy data demonstrate the sensitivity and resolution necessary to monitor these subtle changes in equilibria, even if these are only a few Angstroms. For several of the conditions we investigated by DEER in solution, corresponding x-ray structures have been solved, with the derived distances agreeing well with the DEER distributions. This further validates the biological relevance of the structures, including serial time-resolved ones that indicate asymmetry.

The ZLD-bound crystal structure doesn't predict the DEER distances, and the conformation of Na⁺ binding site sidechains in the ZLD structure doesn't predict whether sodium currents occur. This might suggest that the ZLD structure captures a conformation that does not recapitulate what is happening in solution/ a membrane.

We agree with the reviewer that the ZLD-bound crystal structure does not predict the DEER distances. However, we believe this discrepancy arises from the effect of the bulkiness of ZLD inhibitor, which prevents the closure of the hydrolytic centre. Additionally, the absence of Na⁺ at the ion gate in the ZLD-bound structure suggests that Na⁺ transport does not occur, a conclusion further supported by our electrometric measurements. We agree with the reviewer, that the distances observed in the DEER experiments might represent a potential new conformation in solution, which may not be captured by the static X-ray structure, thereby offering insights into the dynamic nature of the protein under physiological conditions. Finally, the static x-ray structures have not captured the asymmetric conformations that must exist to explain half-of-the-sites reactivity.

Reviewer #2 (Public review):

Summary:

Crystallographic analysis revealed the asymmetric conformation of the dimer in the inhibitor-bound state. Based on this result, which is consistent with previous time-resolved analysis, authors verified the dynamics and distance between spin introduced label by DEER spectroscopy in solution and predicted possible patterns of asymmetric dimer.

Strengths:

Crystal structures with inhibitor bound provide detailed coordination in the binding pocket thus useful information for the PPase field and maybe for drug development.

Weaknesses:

The distance information measured by DEER is advantageous for verifying the dynamics and structure of membrane protein in solution. However, regarding T211 data, which, as the authors themselves stated, lacks measurement precision, it is unclear for readers how confident one can judge the conclusion leading from these data for the cytoplasmic side.

We thank the reviewer for acknowledging the advantageous use of the DEER methodology for identifying dynamic states of membrane proteins in solution. We used two sites in our analysis: S525 (periplasm) and T211 (cytoplasm). As we clearly stated in the original manuscript, S525R1 yielded high-quality DEER data, while T211R1 yielded weak (or no) visual oscillations, leading to broad, though different distributions for the several conditions tested. Our main conclusions are based on the S525R1 data. We included the T211R1 data because, although it does not provide definitive evidence, it is consistent with our proposed model and offers additional insights into biologically relevant conditions. Furthermore, the shifts in the centre of mass (Fig EV8D) of the broad T211R1 distributions show a trend that is consistent with our model; although not proving it, it does not exclude it either. Lastly, these data do indeed confirm an important structural feature of mPPase in solution conditions which is the intrinsically high dynamic state of the loop5-6 where T211 is located, and consistent with our previous (Kellosalo *et al.*, Science, 2012; Li *et al.*, Nat. Commun, 2016; Vidilaseris *et al.*, Sci. Adv., 2019; Strauss *et al.*, EMBO Rep., 2024) and current x-ray crystallography data.

The distance information for the luminal site, which the authors claim is more accurate, does not indicate either the possibility or the basis for why it is the ensemble of two components and not simply a structure with a shorter distance than the crystal structure.

We thank the reviewer for pointing out this possibility and alternative interpretation of our DEER data. In the revised version, we will show that our DEER data are consistent with (and do not exclude) asymmetry and rephrase to be inclusive of other possibilities. Importantly, this additional possibility does not affect the current interpretation of the data in our manuscript.

Reviewer #3 (Public review):

Summary:

Membrane-bound pyrophosphatases (mPPases) are homodimeric proteins that hydrolyze pyrophosphate and pump H⁺/Na⁺ across membranes. They are attractive drug targets against protist pathogens. Non-hydrolysable PPi analogue bisphosphonates such as risedronate (RSD) and pamidronate (PMD) serve as primary drugs currently used. Bisphosphonates have a P-C-P bond, with its central carbon can accommodate up to two substituents, allowing a large compound variability. Here the authors solved two TmPPase structures in complex with the bisphosphonates etidronate (ETD) and zoledronate (ZLD) and monitored their conformational ensemble using DEER spectroscopy in solution. These results reveal the inhibition mechanism of these compounds, which is crucial for developing future small molecule inhibitors.

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The authors show that seven different bisphosphonates can inhibit TmPPase with IC50 values in the micromolar range. Branched aliphatic and aromatic modifications showed weaker inhibition.

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The conformational heterogeneity is further investigated using DEER spectroscopy under several conditions.

Weaknesses:

The authors observed asymmetry in the TmPPase-ELD structure above the hydrolytic center. The structural asymmetry arises due to differences in the orientation of ETD within each monomer at the active site. As a result, loop5-6 of the two monomers is oriented differently, resulting in the observed asymmetry. The authors attempt to further establish this asymmetry using DEER spectroscopy experiments. However, the (over)interpretation of these data leads to more confusion than any further understanding. DEER data suggest that the asymmetry observed in the TmPPase-ELD structure in this region might be funneled from the broad conformational space under the crystallization conditions.

See also the response below - We respectfully disagree with the reviewer. The asymmetry was previously established using serial time crystallography (Strauss et al., EMBO Rep, 2024) and biochemical assays (e.g. Malinen et al., Prot. Sci., 2022; Artukka et al., Biochem J, 2018; Luoto et al., PNAS, 2013) and also partially seen in one static structure (Vidilaseris et al., Sci Adv 2019). DEER data only show that the previously proposed asymmetry could also be present within the conformational ensemble in solution conditions. Indeed, our data do not (and cannot) exclude this possibility.

DEER data for position T211R1 at the enzyme entrance reveal a highly flexible conformation of loop5-6 (and do not provide any direct evidence for asymmetry, Figure EV8).

Please see relevant response above. We acknowledge that T211 is indeed situated on a highly dynamic loop, which is important for gating and our DEER data confirm its high flexibility. Given we have not observed oscillations of this site, leading to broad distributions, we have stated in the original manuscript that we will not establish the presence of any asymmetry in solution on the basis of T211, rather relying on the S525 site, for which we have acquired high-quality DEER data, as was also pointed out and have been commented on by all reviewers.

Similarly, data for position S521R1 near the exit channel do not directly support the proposed asymmetry for ETD.

The reviewer appears to suggest that we hold the S525R1 DEER data as direct proof of asymmetry; this is combative on the grounds that to directly prove asymmetry would require time-resolved DEER measurements, far beyond the scope of this work. Rather, we have applied DEER measurements to explore whether asymmetry (observed previously via time-resolved X-ray crystallography) is also present (or indeed a possibility) in solution. We simply state that the DEER data are consistent with asymmetry (i.e., that the mean distance increases in the presence of ETD compared to the apo-state). This is a restrained interpretation of the data.

Despite the high quality of the data, they reveal a very similar distance distribution. The reported changes in distances are very small (± 0.3 nm), which can be accommodated by a change of spin label rotamer distribution alone. Further, these spin labels are located on a flexible loop, thereby making it difficult to directly relate any distance changes to the global conformation

We thank the reviewer for recognising the high quality of our DEER data for the S525R1, where visual oscillations in the raw traces, as in our case, reportedly lead to highly accurate and reliable distributions, able to separate (in fortuitous cases) helical movements of only a few Angstroms. The ability of DEER/PELDOR offering near Angstrom resolution was previously demonstrated by the acquisition and solution of high resolution multi-subunit spin-labelled membrane protein structures (Pliotas *et al.*, PNAS, 2012; Pliotas *et al.*, Nat Struct Mol Biol, 2015; Pliotas, Methods Enzymol, 2017) as well as its ability in detecting small (and of similar to mPPase magnitude) conformational changes in different integral membrane proteins systems (Kapsalis *et al.*, Nature Comm, 2019; Kubatova *et al.*, PNAS, 2023; Schmidt *et al.*, JACS, 2024; Lane *et al.*, Structure, 2024; Hett *et al.*, JACS, 2021; Zhao *et al.*, Nature, 2024), occurring under different conditions and/or stimuli in solution and/or lipid environment. The changes here are not very small (e.g. ~ 7 Angstroms between the two mean distance extremes (Ca vs IDP)) for DEER's proven detection sensitivity, and with all other conditions showing changes between those extremes.

These changes are relatively small, but they are expected for membrane ion pumps. *Indeed, none of the mPPase structures show helical movements of greater than a half a turn, and that only in helices 6 and 12.* There appear to be larger-scale loop closing motions of the 5-6 loop that includes T211, due to the presence of E217 which binds to one of the Mg^{2+} ions that coordinate the leaving group phosphate. (This is, *inter alia*, the reason that this loop is so flexible: it can not order before substrate is bound.) Here we have the resolution to detect such subtle differences by DEER, given there are clear shifts in our time domain data and these are reflected in the mean distances in the distributions. Therefore, our study demonstrates the sensitivity and resolution DEER offers in detecting subtle conformational transitions, key in membrane proteins pathways. To further belabour this point, we do not quantify the DEER data (for instance through parametric fitting) to extract populations of different conformational states and we appreciate that to do so would be highly prone to error; however we do (and can, we feel without overinterpretation) assert that the mean distances shift.

The interpretations listed below are not supported by the data presented:

(1) 'In the presence of Ca^{2+} , the distance distribution shifts towards shorter distances, suggesting that the two monomers come closer at the periplasmic side, and consistent with the predicted distances derived from the TmPPase:Ca structure.' Problem: This is a far-stretched interpretation of a tiny change, which is not reliable for the reasons described in the paragraph above.

While the authors overall agree with the reviewer assessment that ± 0.3 nm is a small (not a minor) change, there are literature examples quantifying (or using for quantification) distribution peaks separated by similar Δr . (Kubatova *et al.*, PNAS, 2023; Schmidt *et al.*, JACS, 2024; Hett *et al.*, JACS, 2021; Zhao *et al.*, Nature, 2024). In particular, none of the mPPase structures show helical movements of greater than a half a turn (in helices 6 and 12 in particular). There appear to be larger-scale loop closing motions of the 5-6 loop that includes T211, due to the presence of E217 which binds to one of the Mg^{2+} ions that coordinate the leaving group phosphate. (This is, *inter alia*, the reason that this loop is so flexible: it can not order before substrate is bound.)

Importantly, we have fitted Gaussians to the experimental distance distributions of 525R1 output by the Comparative Deer Analyzer 2.0 and observed a change in the distribution width in presence of Ca²⁺, implying the rotameric freedom of the spin label is restricted. However, the CW-EPR for 525R1 indicate that the rotational correlation time of the spin label is highly consistent between conditions (the spectra are almost identical); this cannot be explained simply by rotameric preference of the spin label (as asserted by the reviewer 3), as there is no (further) immobilisation observed from the CW-EPR of apo-state (Figure EV9) to that in presence of Ca²⁺. Furthermore, in the absence of conformational changes, it is reasonable to assume (and demonstrable from the CW-EPR data) that the rotamer cloud should not significantly change between conditions. However, Gaussian fits of the two extreme cases yielding the longest (i.e., in presence of IDP) and shortest (in presence of ZTD) mean distances for the 525R1 DEER data indicated significant (i.e., above the noise floor after Tikhonov validation) probability density for the IDP condition at 50 Å ($P(r) = 0.18$). This occurs at four standard deviations above the mean of the ZTD condition, which by random chance should occur with <0.007% probability. Indeed, one can say that to observe 18% probability density at four standard deviations above the mean by random chance would occur on the order of one in 4×10^6 .

As in previous response the method can detect changes of such magnitude which are not small, but physiologically relevant and expected for integral membrane proteins, such as mPPases. Indeed, even in equal (or more) complex systems such as heptameric mechanosensitive channel proteins DEER provided sub-Angstrom accuracy, when a spin labelled high resolution XRC structure was solved (Pliotas et al., PNAS, 2012; Pliotas et al., Nat Struct Mol Biol, 2015). Despite this is ideal case where DEER accuracy was experimentally validated another high resolution structural method on modified membrane protein and is not very common it demonstrates the power of the method, especially when strong oscillations are present in the raw DEER data (as here for mPPase 525R1), even when multiple distances are present, Angstrom resolution is achievable in such challenging protein classes.

(2) 'Based on the DEER data on the IDP-bound TmPPase, we observed significant deviations between the experimental and the in silico distances derived from the TmPPase:IDP X-ray structure for both cytoplasmic- (T211R1) and periplasmic-end (S525R1) sites (Figure 4D and Figure EV8D). This deviation could be explained by the dimer adopting an asymmetric conformation under the physiological conditions used for DEER, with one monomer in a closed state and the other in an open state.'

Problem: The authors are trying to establish asymmetry using the DEER data. Unfortunately, no significant difference is observed (between simulation and experiment) for position 525 as the authors claim (Figure 4D bottom panel). The observed difference for position 112 must be accounted for by the flexibility and the data provide no direct evidence for any asymmetry.

Reviewer 3 is wrong in suggesting that we are trying to prove asymmetry through the DEER data. That is a well-known fact in the literature (eg Vidilaseris et al, Sci Adv 2019 where we show (1) that the exit channel inhibitor ATC (i.e., close to 525) binds better in solution to the TmPPase:PPi complex than the TmPPase:PPi2 complex, and (2) that ATC binds in an asymmetric fashion to the TmPPase:IDP2 complex with just one ATC dimer on one of the exit channels. We merely use the DEER data to support this well-established fact.

However, we agree that the DEER data in presence of IDP does not provide direct proof for asymmetry; particularly mutant T211R1 yields in silico distributions too short for measurement by DEER. It is possible that the deviations observed (and particularly likely for T211R1) arise from conformational heterogeneity in solution. We will rephrase this paragraph accordingly: "Owing to the broad nature of the T211R1 (cytoplasmic site) distance

distributions, we refrain from interpreting shifts in this data. For the 525R1 (periplasmic site) for which we obtained data of high quality (as also pointed out by both reviewers 2 and 3) we observed deviations between the experimental and the *in-silico* distances derived from the TmPPase:IDP X-ray structure. While this deviation is less pronounced than for the +ZTD condition, the deviation is consistent with an asymmetric conformation in solution.”

(3) *'Our new structures, together with DEER distance measurements that monitor the conformational ensemble equilibrium of TmPPase in solution, provide further solid experimental evidence of asymmetry in gating and transitional changes upon substrate/inhibitor binding.'*

Problem: See above. The DEER data do not support any asymmetry.

We feel that the reviewer comments here are somewhat unfounded. The DEER data (and we will limit discussion only to the 525R1 mutant in this regard) satisfy relevant criteria of the white paper (Schiemann et al., 2021, JACS) from the EPR community (signal-to-noise ratio w.r.t modulation depth of > 20 in all cases; replicates have been performed and will be added into the main-text or supplementary; near quantitative labelling efficiency (evidenced by lack of free spin label signal in the CW-EPR spectra); analysed using the CDA (now Figure EV10, this data we will promote to the main-text) to avoid confirmation bias).

While the DEER data do not prove asymmetry, we do not claim proof of asymmetry in the above sentence. We concede to rephrase the offending sentence above as: “Our new structures, together with DEER distance measurements that monitor the conformational ensemble of TmPPase in solution, do not exclude asymmetry in gating and transitional changes upon substrate/inhibitor binding and are consistent with our proposed model.” We feel that this reframed conjecture of asymmetry is well founded; indeed, comparing the experimental apo-state 525R1 distance distribution with *in-silico* modelling performed on the hybridised asymmetric structure (i.e., comprised of one monomer bound to Ca²⁺ and another bound to IDP) yields an overlap coefficient (Islam and Roux, JPC B, 2015) of >0.97. This implies the envelope of the modelled distance distribution is quantitatively inside the envelope of the experimental distance distribution. Thus, the DEER data do not exclude asymmetry (previously observed by time-resolved XRC) in solution. While we appreciate that ideally one would measure time-resolved DEER to directly correlate kinetics of conformational changes within the ensemble to the catalytic cycle of mPPase, (and this is something we aim to do in the future), it is beyond the the scope of this study.

Indeed, half-of-the-sites reactivity has been demonstrated in at least the following papers (Vidilaseris et al, Sci Acv. ,2019, Strauss et al, EMBO Rep. 2024, Malinen et al Prot Sci, 2022, Artukka et al Biochem J, 2018; Luoto et al, PNAS, 2013). Half-of-the sites activity requires asymmetry in the mechanism, and therefore asymmetric motions in the active site (viz 211) and exit channel (viz 525). As mentioned above, we have demonstrated this for other inhibitors (Vidilaseris et al 2019) and as part of a time-resolved experiment (Strauss et al 2024). In fact, given the wealth of evidence showing that the symmetrical crystal structures sample a non- or less-productive conformation of the protein, it would be quixotic to propose the DEER experiments - in solution - do not generate asymmetric conformations. It certainly doesn't obey Occam's razor of choosing the simplest possible explanation that covers the data.

(4) *Based on these observations, and the DEER data for +IDP, which is consistent with an asymmetric conformation of TmPPase being present in solution, we propose five distinct models of TmPPase (Figure 7).*

Problem: Again, the DEER data do not support any asymmetry and the authors may revisit the proposed models.

We respectfully disagree with the reviewer. Please see our detailed response above. However, in the revised version, we will clarify that the proposed models are not solely based on the DEER data but are grounded in both current and previously solved structures, with the DEER data providing additional consistency with these models.

(5) 'In model 2 (Figure 7), one active site is semi-closed, while the other remains open. This is supported by the distance distributions for S525R1 and T211R1 for +Ca/ETD informed by DEER, which agrees with the in silico distance predictions generated by the asymmetric TmPPase:ETD X-ray structure'

Problem: Neither convincing nor supported by the data

We respectfully disagree with the reviewer. However, owing to the conformational heterogeneity of T211R1, in the revised version, we will exclude it in the above sentence, to the effect: Please see our detailed response above.

<https://doi.org/10.7554/eLife.102288.1.sa0>